

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414599
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Nussbaum; Jonathan

Lace adjuster assembly including feedback assembly for use in visualizing and measuring athletic performance

Abstract

A lace adjuster system for selectively adjusting shoelaces (12) of shoes (10A, 10B) of a user, including a first lace adjuster assembly (13) for use with the first shoe (10A) and a second lace adjuster assembly (13) for use with the second shoe (10B). Each lace adjuster assembly (13) includes (i) a lace adjuster (14); and (ii) a feedback assembly (15) that is mechanically coupled to the lace adjuster (14), the feedback assembly (15) selectively measuring statistical data of the user during an athletic performance, the feedback assembly (15) including a sensor assembly (216) that senses a performance characteristic of the user during the athletic performance; and a controller (360) that receives the performance characteristic and generates a statistical data point that is based at least in part on the performance characteristic. The statistical data points can be combined to generate a combined statistical data point having enhanced accuracy.

Inventors:	Nussbaum; Jonathan (Miami Beach, FL)
Applicant:	LACECLIP LLC (Miami Beach, FL)
Family ID:	1000008820092
Assignee:	LACECLIP LLC (Miami Beach, FL)
Appl. No.:	18/444515
Filed:	February 16, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240277105 A1	Aug. 22, 2024

Related U.S. Application Data

continuation parent-doc US 16690908 20191121 PENDING child-doc US 18110817
continuation parent-doc US 15301946 US 10595581 20200324 WO PCT/US2015/025763
20150414 child-doc US 16690908
continuation-in-part parent-doc US 18110817 20230216 US 12144397 child-doc US 18444515
continuation-in-part parent-doc US 17354655 20210622 US 11937666 20230530 child-doc US
18444515
us-provisional-application US 63446546 20230217
us-provisional-application US 63042401 20200622
us-provisional-application US 62043822 20140829
us-provisional-application US 62018194 20140627
us-provisional-application US 61979491 20140414

Publication Classification

Int. Cl.: **A43B3/34** (20220101); **A43C1/06** (20060101); **A43C7/00** (20060101); **A43C7/08**
(20060101); **A45F5/02** (20060101); **G01P15/18** (20130101); **G03B29/00** (20210101)

U.S. Cl.:

CPC **A43B3/34** (20220101); **A43C1/06** (20130101); **A43C7/00** (20130101); **A43C7/08**
(20130101); **A45F5/02** (20130101); **G01P15/18** (20130101); **G03B29/00** (20130101);
A45F2005/023 (20130101)

Field of Classification Search

CPC: A43B (3/34); A43C (1/06); A43C (7/00); A43C (7/08); A45F (5/02); A45F (2005/023)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1879475	12/1931	Poon	24/712.9	A43C 7/08
4622723	12/1985	Krauss	24/136R	F16G 11/101
5029371	12/1990	Rosenblood	N/A	N/A
5345657	12/1993	Shimizu	24/136L	F16G 11/101
5649340	12/1996	Ida	24/136R	F16G 11/101
5899963	12/1998	Hutchings	702/145	A61B 5/1123
6876947	12/2004	Darley	702/144	G01P 1/023
7255468	12/2006	Capriola	36/137	A43B 1/0027
7404263	12/2007	Guzman	N/A	N/A
7596891	12/2008	Carnes	36/25R	A43B 3/0031
7695154	12/2009	Ellenburg	362/205	A43B 1/0036
7722219	12/2009	Hartley	362/253	A43B 3/34
7908774	12/2010	Mirza	36/137	A43B 3/24
7927253	12/2010	Vincent	482/8	A63B 24/0087
8172722	12/2011	Molyneux	482/901	A43B 1/0054
8181320	12/2011	Wolfberg	36/50.1	A43C 7/00

8231506	12/2011	Molyneux	482/901	A43B 5/02
8360904	12/2012	Oleson	482/8	A43B 13/00
D689495	12/2012	Bentley et al.	N/A	N/A
D694842	12/2012	Bentley et al.	N/A	N/A
8613676	12/2012	Bentley	N/A	N/A
8628453	12/2013	Balakrishnan	482/901	H04N 23/66
8700354	12/2013	Bentley et al.	N/A	N/A
8702430	12/2013	Dibenedetto	434/258	H04M 1/72412
8702516	12/2013	Bentley	463/40	G06T 7/20
D706654	12/2013	Bentley et al.	N/A	N/A
8739639	12/2013	Owings	73/862.391	A61B 5/1036
8769836	12/2013	Donovan	33/3A	A43B 3/38
8827824	12/2013	Bentley et al.	N/A	N/A
8903521	12/2013	Goree et al.	N/A	N/A
8904605	12/2013	Kawaguchi	D8/383	F16G 11/03
8905855	12/2013	Fitzpatrick et al.	N/A	N/A
8913134	12/2013	Goree et al.	N/A	N/A
8941723	12/2014	Bentley	348/47	A61B 5/6895
8944928	12/2014	Kaps et al.	N/A	N/A
8994826	12/2014	Bentley	473/223	G01S 19/19
9017296	12/2014	Beck	604/249	A61M 39/28
9028337	12/2014	Bentley	N/A	N/A
9039527	12/2014	Bentley et al.	N/A	N/A
9301569	12/2015	Donovan	N/A	A43B 3/36
9542706	12/2016	Case, Jr.	N/A	G08B 1/08
9642415	12/2016	Pease	N/A	N/A
9743861	12/2016	Giedwoyn	N/A	G16H 20/30
9763489	12/2016	Amos	N/A	A63F 13/214
9928878	12/2017	Anderson	N/A	N/A
10238166	12/2018	Rosenbaum	N/A	N/A
10363453	12/2018	Fitzgerald	N/A	A43B 3/34
10409961	12/2018	Flaherty	N/A	N/A
10512304	12/2018	Nussbaum	N/A	A43C 7/08
10595581	12/2019	Nussbaum	N/A	A45F 5/02
10667580	12/2019	Kovach et al.	N/A	N/A
10926133	12/2020	Giedwoyn	N/A	A63B 24/0021
10983945	12/2020	Molettiere	N/A	G16H 10/60
11060926	12/2020	Reif	N/A	G16H 50/20
11510462	12/2021	Yun	N/A	A43C 11/20
11603905	12/2022	Perrier	N/A	F16G 11/101
11662359	12/2022	Kolen	73/379.01	A61B 5/11
11684111	12/2022	Henrichot	700/275	A43B 3/34
11937666	12/2023	Nussbaum	N/A	A45F 5/02
2002/0152645	12/2001	Darley et al.	N/A	N/A
2004/0221433	12/2003	Wolfberg	24/712.5	A43B 3/34
2005/0268436	12/2004	Yoshiguchi	24/115G	F16G 11/101
2007/0166024	12/2006	Robinson	N/A	N/A
2007/0260421	12/2006	Berner, Jr.	702/155	A43B 13/00

2008/0115334	12/2007	Chen	24/115G	A43C 7/00
2009/0223085	12/2008	Wolfberg	N/A	N/A
2009/0293313	12/2008	Bruce et al.	N/A	N/A
2011/0016680	12/2010	Chang	24/712.5	A43C 7/00
2011/0178720	12/2010	Bamburg et al.	N/A	N/A
2011/0251822	12/2010	Darley et al.	N/A	N/A
2012/0088544	12/2011	Bentley et al.	N/A	N/A
2012/0215474	12/2011	Bentley et al.	N/A	N/A
2013/0008058	12/2012	Jasmine	36/136	A43B 3/34
2013/0095941	12/2012	Bentley et al.	N/A	N/A
2013/0117975	12/2012	Herrera	N/A	N/A
2013/0128022	12/2012	Bose et al.	N/A	N/A
2013/0185003	12/2012	Carbeck	702/41	G01L 1/00
2013/0211774	12/2012	Bentley et al.	N/A	N/A
2014/0085465	12/2013	Angermann	348/143	G01C 21/005
2014/0376876	12/2013	Bentley et al.	N/A	N/A
2015/0059136	12/2014	Schreiner	24/712.3	A43C 7/00
2015/0154452	12/2014	Bentley et al.	N/A	N/A
2017/0241797	12/2016	Kong et al.	N/A	N/A
2020/0008745	12/2019	Burch, V	N/A	A61B 5/0004
2020/0375470	12/2019	Fu	N/A	A43B 7/147
2021/0012877	12/2020	Aragones et al.	N/A	N/A
2021/0014582	12/2020	Case, Jr.	N/A	N/A
2021/0315315	12/2020	Nussbaum	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101185535	12/2007	CN	N/A
101367011	12/2008	CN	N/A
2009050699	12/2008	JP	N/A
2014504943	12/2013	JP	N/A
20010065023	12/2000	KR	N/A
WO2013010171	12/2012	WO	N/A
WO2013184672	12/2012	WO	N/A
WO2014036374	12/2013	WO	N/A
WO2015191157	12/2014	WO	N/A
WO2015191157	12/2014	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion, PCT Application Serial No.

PCT/US2024/016203, dated Jun. 26, 2024, issued by International Search Authority US. cited by applicant

Decision on Application issued on Oct. 17, 2023, by the China National Intellectual Property Administration, in CNIPA Application Serial No. 2021111964284. Informal English translation with redactions provided. cited by applicant

The International Search Report and Written Opinion of the International Searching Authority, dated Jul. 20, 2015, FlyClip LLC, PCT/US15/25763. (related application). cited by applicant

The International Preliminary Report on Patentability, dated Oct. 18, 2016, FlyClip LLC, PCT/US15/25763. (related application). cited by applicant

“Beyond Fitbit: Sensors track hang time, vertical leaps”, San Diego Union Tribune Newspaper Article by Mike Freeman dated Jun. 17, 2015, pp. 1-3. cited by applicant
Supplementary European Search Report from the European Patent Office, U.S. Appl. No. 15/301,946, dated Oct. 16, 2017. cited by applicant
Office Action issued on Feb. 17, 2018 by the State Intellectual Property Office of China in Application No. 2015800318712 (this Chinese application is related to the instant U.S. Application and both claim priority from PCT application PCT/US2015/025763). cited by applicant
Office Action issued on Oct. 17, 2018 by the State Intellectual Property Office of China in Application No. 2015800318712 (this Chinese application is related to the instant U.S. Application and both claim priority from PCT application PCT/US2015/025763). cited by applicant
Office Action issued on Oct. 15, 2018 by the European Patent Office in Application No. 15806231.5 (this EPO application is related to the instant U.S. Application and both claim priority from PCT application PCT/US2015/025763). cited by applicant
Response to Office Action filed on Feb. 14, 2019 with the European Patent Office in Application No. 15806231.5 (this EPO application is related to the instant U.S. Application and both claim priority from PCT application PCT/US2015/025763). cited by applicant
Office Action issued on Mar. 4, 2019 by the Australian Intellectual Property Office in Application No. 2015272024 (this Australian application is related to the instant U.S. Application and both claim priority from PCT application PCT/US2015/025763). cited by applicant
Office Action issued on Mar. 26, 2019 by the Japan Patent Office in Application No. 2016-562003 (this Japanese application is related to the instant U.S. Application and both claim priority from PCT application PCT/US2015/025763). cited by applicant
Office Action issued on Oct. 29, 2019 by the Japan Patent Office in Application No. 2016-562003 (this Japanese application is related to the instant U.S. Application). cited by applicant
Office Action issued by the U.S. Patent Office in U.S. Appl. No. 17/354,655 dated Feb. 21, 2023. cited by applicant
First Office Action issued by the Canadian Intellectual Property Office on Apr. 1, 2020, in Canadian Patent Application No. 2,945,792. (This application is related to U.S. Appl. No. 16/690,908). cited by applicant
Office Action from the State Intellectual Property Office—China, in Application No. 2015800318712, dated Apr. 1, 2021. (This application is related to U.S. Appl. No. 16/690,908). cited by applicant
International Search Report and Written Opinion, issued by European Patent Office, in Application No. PCT/2021/038485, dated Oct. 12, 2021. (Related matter). cited by applicant
Office Action issued by the China Intellectual Property Administration, on Nov. 30, 2022, in Chinese Patent Application No. 2021111964284. (Related matter). cited by applicant
Notification Concerning Transmittal of International Preliminary Report on Patentability issued by the International Bureau on Jan. 5, 2023, in PCT patent application serial No. PCT/US2021/038485. (Related matter). cited by applicant

Primary Examiner: San; Jason W

Assistant Examiner: Sullivan; Matthew J

Attorney, Agent or Firm: ROEDER & BRODER LLP

Background/Summary

RELATED APPLICATIONS (1) The present application claims priority on U.S. Provisional Application Ser. No. 63/446,546, filed on Feb. 17, 2023, and entitled “LACE ADJUSTER ASSEMBLY INCLUDING FEEDBACK ASSEMBLY FOR USE IN VISUALIZING AND MEASURING ATHLETIC PERFORMANCE”. As far as permitted, the contents of U.S. Provisional Application Ser. No. 63/446,546 are incorporated in their entirety herein by reference. (2) Additionally, the present application is a continuation-in-part application of U.S. application Ser. No. 18/110,817 filed on Feb. 16, 2023, and entitled “LACE ADJUSTER ASSEMBLY INCLUDING FEEDBACK ASSEMBLY FOR USE IN VISUALIZING AND MEASURING ATHLETIC PERFORMANCE”. U.S. application Ser. No. 18/110,817 is a continuation application of U.S. application Ser. No. 16/690,908 filed on Nov. 21, 2019, and entitled “LACE ADJUSTER ASSEMBLY INCLUDING FEEDBACK ASSEMBLY FOR USE IN VISUALIZING AND MEASURING ATHLETIC PERFORMANCE”; which is a continuation application of U.S. application Ser. No. 15/301,946 filed on Oct. 4, 2016 (now U.S. Pat. No. 10,595,581 B2 issued on Mar. 24, 2020), and entitled “LACE ADJUSTER ASSEMBLY INCLUDING FEEDBACK ASSEMBLY FOR USE IN VISUALIZING AND MEASURING ATHLETIC PERFORMANCE”. As far as permitted, the contents of U.S. application Ser. Nos. 18/110,817, 16/690,908, and 15/301,946 (now U.S. Pat. No. 10,595,581 B2) are incorporated in their entirety herein by reference. (3) U.S. application Ser. No. 15/301,946 is a 371 of PCT/US2015/025763 filed on Apr. 14, 2015, and entitled “LACE ADJUSTER ASSEMBLY INCLUDING FEEDBACK ASSEMBLY FOR USE IN VISUALIZING AND MEASURING ATHLETIC PERFORMANCE”. PCT Application Serial No: PCT/US2015/025763 is related to and claims priority on (i) U.S. Provisional Application Ser. No. 61/979,491 filed on Apr. 14, 2014, and entitled “LACE ADJUSTER”; (ii) U.S. Provisional Application Ser. No. 62/018,194 filed on Jun. 27, 2014, and entitled “SENSOR ASSEMBLY FOR USE IN MEASURING ATHLETIC PERFORMANCE”; and (iii) U.S. Provisional Application Ser. No. 62/043,822 filed on Aug. 29, 2014, and entitled “IMAGE ASSEMBLY AND SENSOR ASSEMBLY FOR USE IN VISUALIZING AND MEASURING ATHLETIC PERFORMANCE”. To the extent permissible, the contents of (i) PCT Application Serial No: PCT/US2015/025763, and (ii) U.S. Provisional Application Ser. Nos. 61/979,491, 62/018,194, and 62/043,822 are incorporated in their entirety herein by reference. (4) Further, the present application is also a continuation-in-part application of U.S. application Ser. No. 17/354,655 filed on Jun. 22, 2021, and entitled “LACE ADJUSTER ASSEMBLY INCLUDING FEEDBACK ASSEMBLY FOR USE IN VISUALIZING AND MEASURING ATHLETIC PERFORMANCE”. U.S. application Ser. No. 17/354,655 is related to and claims priority on U.S. Provisional Application Ser. No. 63/042,401 filed on Jun. 22, 2020, and entitled “LACE ADJUSTER”. To the extent permissible, the contents of U.S. application Ser. No. 17/354,655, and U.S. Provisional Application Ser. No. 63/042,401 are incorporated in their entirety herein by reference.

BACKGROUND

(1) Many athletes, professional or amateur, serious, or casual, are very interested in visualizing, quantifying and/or improving their athletic performances. Thus, it is desired to provide a device that enables such athletes to effectively visualize and/or gauge various aspects of their athletic performance through generation of performance metrics and statistical data, which can then be subsequently used as a means to view unique perspectives of their athletic performance and/or to improve their athletic performance over time.

(2) Additionally, it is often necessary to adjust, tighten, and untighten (or loosen) the shoelaces of a shoe. It is further desired to inhibit the shoelaces from being a potential tripping hazard for the person wearing the shoes. This can be especially true for an athlete during an athletic performance, as problems with shoelaces being untied, too tight, or too loose, and/or becoming tripping hazards, can lead to suboptimal performance and/or injury.

SUMMARY

(3) The present invention is directed toward a lace adjuster system that is adapted to selectively adjust a first shoelace of a first shoe of a user and to selectively adjust a second shoelace of a second shoe of the user. In various embodiments, the lace adjuster system includes a first lace adjuster assembly and a second lace adjuster assembly. The first lace adjuster assembly includes (i) a first lace adjuster that is adapted to selectively adjust the first shoelace of the first shoe of the user; and (ii) a first feedback assembly that is mechanically coupled to the first lace adjuster, the first feedback assembly being configured to selectively measure statistical data of the user during an athletic performance, the first feedback assembly including a first sensor assembly including a first performance sensor that is mechanically coupled to the first lace adjuster and that senses a first performance characteristic of the user during the athletic performance; and a first controller that is electrically coupled to the first performance sensor, the first controller including a first processor, the first controller receiving the first performance characteristic from the first performance sensor and generating a first statistical data point that is based at least in part on the first performance characteristic. The second lace adjuster assembly includes (i) a second lace adjuster that is adapted to selectively adjust the second shoelace of the second shoe of the user; and (ii) a second feedback assembly that is mechanically coupled to the second lace adjuster, the second feedback assembly being configured to selectively measure statistical data of the user during the athletic performance, the second feedback assembly including a second sensor assembly including a second performance sensor that is mechanically coupled to the second lace adjuster and that senses a second performance characteristic of the user during the athletic performance; and a second controller that is electrically coupled to the second performance sensor, the second controller including a second processor, the second controller receiving the second performance characteristic from the second performance sensor and generating a second statistical data point that is based at least in part on the second performance characteristic.

(4) In some embodiments, the first controller is the same as the second controller.

(5) In certain embodiments, the first controller and the second controller are each within a remote device.

(6) In many embodiments, the first statistical data point and the second statistical data point are combined by one of the first controller and the second controller to generate a combined statistical data point having enhanced accuracy relative to the first statistical data point and the second statistical data point.

(7) In certain embodiments, the first feedback assembly further includes a first storage device for storing the combined statistical data point.

(8) In some embodiments, the first storage device is mechanically coupled to the lace adjuster; and the first sensor assembly further includes a first transmitter for transmitting the combined statistical data point from the first storage device to a remote device.

(9) In certain embodiments, the first transmitter transmits the combined statistical data point from the first storage device to the remote device via a wireless connection. In other embodiments, the first transmitter transmits the combined statistical data point from the first storage device to the remote device via a wired connection.

(10) In various embodiments, the first storage device is positioned within a remote device.

(11) In many embodiments, the first performance sensor senses one or more of a horizontal movement, a vertical movement and an angular movement of the user during the athletic performance.

(12) In some embodiments, the first performance sensor is one of a two-axis accelerometer, a three-axis accelerometer, and a three-axis gyrometer.

(13) In certain embodiments, the first performance sensor includes a first magnetometer that measures a magnitude and direction of magnetic fields at a point in space in relation to a position of the user during the athletic performance.

- (14) In some embodiments, the second performance sensor senses the one or more of a horizontal movement, a vertical movement and an angular movement of the user during the athletic performance.
- (15) In certain embodiments, the second performance sensor is the one of a two-axis accelerometer, a three-axis accelerometer, and a three-axis gyrometer.
- (16) In some embodiments, the second performance sensor includes a second magnetometer that measures a magnitude and direction of magnetic fields at a point in space in relation to a position of the user during the athletic performance.
- (17) In many embodiments, the first feedback assembly further includes a first image capturing assembly that captures a first image of the user during the athletic performance.
- (18) In various embodiments, the second feedback assembly further includes a second image capturing assembly that captures a second image of the user during the athletic performance.
- (19) In certain embodiments, the first image capturing assembly includes a first optical assembly and a first capturing system, and the first optical assembly focuses light onto the first capturing system so that the first capturing system can capture the first image of the user.
- (20) In some embodiments, the first image of the user is one of a still image and a video image.
- (21) In certain embodiments, the first sensor assembly further includes a first locational sensor for providing precise locational information of the user; and the locational information from the first locational sensor is wirelessly transmitted to a remote device.
- (22) In some embodiments, the second sensor assembly further includes a second locational sensor for providing precise locational information of the user; and the locational information from the second locational sensor is wirelessly transmitted to the remote device.
- (23) In many embodiments, the first lace adjuster includes (i) a body assembly having a first body member and a second body member that is coupled to the first body member, the body assembly defining a cavity, and (ii) a lace end retainer that is connected to the body assembly, the lace end retainer being configured to selectively retain at least a portion of the shoelace, the lace adjuster being selectively movable between an unlocked configuration and a locked configuration, wherein the shoelace is adjustable relative to the lace adjuster when the lace adjuster is in the unlocked configuration, and wherein the shoelace is resiliently retained by the lace adjuster and is inhibited from being adjusted relative to the lace adjuster when the lace adjuster is in the locked configuration; and wherein the first performance sensor is positioned within the cavity.
- (24) The present invention is further directed toward a lace adjuster system that is adapted to selectively adjust a first shoelace of a first shoe of a user and to selectively adjust a second shoelace of a second shoe of the user, including a first lace adjuster assembly including (i) a first lace adjuster that is adapted to selectively adjust the first shoelace of the first shoe of the user; and (ii) a first feedback assembly that is mechanically coupled to the first lace adjuster, the first feedback assembly being configured to selectively measure statistical data of the user during an athletic performance, the first feedback assembly including a first sensor assembly including a first performance sensor that is mechanically coupled to the first lace adjuster and that senses a first performance characteristic of the user during the athletic performance; and a first controller that is electrically coupled to the first performance sensor, the first controller including a first processor, the first controller receiving the first performance characteristic from the first performance sensor and generating a first statistical data point that is based at least in part on the first performance characteristic; and a second lace adjuster assembly including (i) a second lace adjuster that is adapted to selectively adjust the second shoelace of the second shoe of the user; and (ii) a second feedback assembly that is mechanically coupled to the second lace adjuster, the second feedback assembly being configured to selectively measure statistical data of the user during the athletic performance, the second feedback assembly including a second sensor assembly including a second performance sensor that is mechanically coupled to the second lace adjuster and that senses a second performance characteristic of the user during the athletic performance; and a second

controller that is electrically coupled to the second performance sensor, the second controller including a second processor, the second controller receiving the second performance characteristic from the second performance sensor and generating a second statistical data point that is based at least in part on the second performance characteristic; wherein the first statistical data point and the second statistical data point are combined by one of the first controller and the second controller to generate a combined statistical data point having enhanced accuracy relative to the first statistical data point and the second statistical data point; wherein the first feedback assembly further includes a first storage device for storing the combined statistical data point; wherein the first storage device is mechanically coupled to the first lace adjuster, the first sensor assembly further including a first transmitter for transmitting the combined statistical data point from the first storage device to a remote device; wherein the first performance sensor senses one or more of a horizontal movement, a vertical movement and an angular movement of the user during the athletic performance; wherein the second performance sensor senses the one or more of a horizontal movement, a vertical movement and an angular movement of the user during the athletic performance; wherein the first feedback assembly includes a first image capturing assembly that captures a first image of the user during the athletic performance; wherein the second feedback assembly includes a second image capturing assembly that captures a second image of the user during the athletic performance; wherein the first sensor assembly further includes a first locational sensor for providing precise locational information of the user, the locational information from the first locational sensor being wirelessly transmitted to a remote device; and wherein the second sensor assembly further includes a second locational sensor for providing precise locational information of the user, the locational information from the second locational sensor being wirelessly transmitted to the remote device.

(25) This summary is an overview of some of the teachings of the present application and is not intended to be an exclusive or exhaustive treatment of the present subject matter. Further details are found in the detailed description and appended claims. Other aspects will be apparent to persons skilled in the art upon reading and understanding the following detailed description and viewing the drawings that form a part thereof, each of which is not to be taken in a limiting sense. The scope herein is defined by the appended claims and their legal equivalents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The novel features of this invention, as well as the invention itself, both as to its structure and its operation, will be best understood from the accompanying drawings, taken in conjunction with the accompanying description, in which similar reference characters refer to similar parts, and in which:
- (2) FIG. 1 is a perspective view of an embodiment of a pair of shoes, each shoe including a shoe body and a shoelace, and an embodiment of a lace adjuster assembly having features of the present invention that can be selectively coupled to the shoelace of each of the shoes, the lace adjuster assembly including a lace adjuster, and a feedback assembly, including a sensor assembly and an image capturing assembly, that is mechanically coupled to the lace adjuster;
- (3) FIG. 2A is a front perspective view of the lace adjuster assembly illustrated in FIG. 1, the lace adjuster being shown in an unlocked configuration;
- (4) FIG. 2B is a front perspective view of a portion of the shoelace, and the lace adjuster assembly illustrated in FIG. 2A, the lace adjuster being shown in a locked configuration;
- (5) FIG. 2C is a rear perspective view of the lace adjuster assembly illustrated in FIG. 2A, the lace adjuster again being shown in the unlocked configuration;
- (6) FIG. 2D is a rear perspective view of the lace adjuster assembly illustrated in FIG. 2A, the lace adjuster again being shown in the locked configuration;

- (7) FIG. 2E is a top view of the lace adjuster illustrated in FIG. 2A;
- (8) FIG. 2F is a cutaway view of the lace adjuster assembly taken on line F-F in FIG. 2E, the lace adjuster being shown in the unlocked configuration;
- (9) FIG. 2G is a sectional view of the lace adjuster assembly illustrated in FIG. 2A, the lace adjuster being shown in the locked configuration;
- (10) FIG. 2H is a front perspective view of the lace adjuster assembly illustrated in FIG. 2A;
- (11) FIG. 2I is another front perspective view of the lace adjuster assembly illustrated in FIG. 2A;
- (12) FIG. 2J is still another front perspective view of the lace adjuster assembly illustrated in FIG. 2A;
- (13) FIG. 2K is a rear perspective view of the lace adjuster assembly illustrated in FIG. 2A;
- (14) FIG. 2L is another rear perspective view of the lace adjuster assembly illustrated in FIG. 2A;
- (15) FIG. 2M is a bottom view of the lace adjuster assembly illustrated in FIG. 2A;
- (16) FIG. 2N is another bottom view of the lace adjuster assembly illustrated in FIG. 2A;
- (17) FIG. 2O is a side view of the lace adjuster assembly illustrated in FIG. 2A;
- (18) FIG. 2P is a front view of the lace adjuster assembly illustrated in FIG. 2A;
- (19) FIG. 2Q is another side view of the lace adjuster assembly illustrated in FIG. 2A;
- (20) FIG. 2R is a rear view of the lace adjuster assembly illustrated in FIG. 2A;
- (21) FIG. 3 is a simplified schematic illustration of an embodiment of the sensor assembly;
- (22) FIG. 4 is a simplified schematic illustration of an embodiment of the image capturing assembly; and
- (23) FIG. 5 is a simplified top view of an athletic field that is usable by a user of the lace adjuster.
- (24) While embodiments of the present invention are susceptible to various modifications and alternative forms, specifics thereof have been shown by way of example and drawings, and are described in detail herein. It is understood, however, that the scope herein is not limited to the particular embodiments described. On the contrary, the intention is to cover modifications, equivalents, and alternatives falling within the spirit and scope herein.

DESCRIPTION

- (25) FIG. 1 is a perspective view of an embodiment of a pair of shoes, i.e. a first shoe **10A** and a second shoe **10B**, with each shoe **10A**, **10B** including a shoe body **11** and a shoelace **12** that is coupled to the shoe body **11**, and an embodiment of a lace adjuster assembly **13** having features of the present invention that can be selectively coupled to the shoelace **12** of each of the shoes **10A**, **10B**. In various embodiments, the lace adjuster assembly **13** includes a lace adjuster **14**, and a feedback assembly **15**, including one or more of a sensor assembly **216** (illustrated more clearly, for example, in FIG. 2F) and an image capturing assembly **17** (also referred to herein simply as an “image assembly”), that is mechanically coupled to the lace adjuster **14**. Alternatively, in certain non-exclusive alternative embodiments, the lace adjuster assembly **13** and/or the feedback assembly **15** can be designed without the sensor assembly **216** and/or without the image assembly **17**.
- (26) The shoes **10A**, **10B**, including the shoe body **11** and the shoelace **12**, can have any suitable design, shape and/or size to meet the specific desires and requirements of the user. As illustrated in FIG. 1, the shoes **10A**, **10B** can be athletic-type shoes that can be used by a user for running, walking, engaging in any of various athletic performances, or for any other chosen activity. Alternatively, the shoes **10A**, **10B** can be another type of shoe. As shown in FIG. 1, the shoelace **12** includes a first lace end **18** having a first end tip **18A**, and an opposed second lace end **19** having a second end tip **19A**.
- (27) As an overview, the feedback assembly **15** is uniquely configured to provide statistical data (via the sensor assembly **216**) and images (via the image assembly **17**) to an athlete (also sometimes referred to herein generally as a “user”) who is using the lace adjuster assembly **13** and/or the feedback assembly **15**. In certain implementations, the user can utilize separate lace adjuster assemblies **13**, and thus separate feedback assemblies **15**, on each of their shoes **10A**, **10B**.

More specifically, in such implementations, the user can have and utilize (i) a first lace adjuster assembly **13** that is coupled to a first shoelace **12** of the first shoe **10A**, the first lace adjuster assembly **13** including a first lace adjuster **14** and a first feedback assembly **15** that is mechanically coupled to the first lace adjuster **14**; and (ii) a second lace adjuster assembly **13** that is coupled to a second shoelace **12** of the second shoe **10B**, the second lace adjuster assembly including a second lace adjuster **14** and a second feedback assembly **15** that is mechanically coupled to the second lace adjuster **14**. In such implementations, the first lace adjuster assembly **13** and the second lace adjuster assembly **13** can be collectively referred to as a “lace adjuster system”; and the first feedback assembly **15** and the second feedback assembly **15** can be referred to collectively as a “feedback system”.

(28) It is appreciated that in such embodiments that include two separate lace adjuster assemblies **13** and two separate feedback assemblies **15**, the first shoe **10A** and the second shoe **10B** can be substantially identical with the exception of one shoe being a right shoe and the other shoe being a corresponding left shoe.

(29) It is further appreciated that the separate lace adjuster assemblies **13** and/or the separate feedback assemblies **15** will typically be identical to one another, but just simply attached to different shoes **10A**, **10B**, such as the first (right) shoe **10A** and the second (left) shoe **10B**. Alternatively, the lace adjuster assemblies **13** and/or the feedback assemblies **15** for each shoe **10A**, **10B** can be different from one another in any desired manner.

(30) As described herein, it is appreciated that the sensor assembly **216** is able to provide different, enhanced and more accurate statistical data to the user when the user has a separate lace adjuster assembly **13**, and thus a separate sensor assembly **216**, coupled to the shoelaces **12** of each of their two shoes **10A**, **10B**. The image assembly **17** is also able to provide additional and potentially more unique images for the user when the user has a separate lace adjuster assembly **13** coupled to the shoelaces **12** of each of their two shoes **10A**, **10B**.

(31) In various embodiments, the sensor assembly **216** can be uniquely designed to provide the athlete who is using the sensor assembly **216**, such as in conjunction with the lace adjuster **14**, with sensed performance characteristics from which can be derived statistical data that enables the athlete to effectively gauge various aspects of their athletic performance and/or evaluate biomechanical movements for injury prevention. In different embodiments, the sensor assembly **216** can include one or more performance sensors **216P** (illustrated in FIG. 2F) in order to provide sensed performance characteristics that are usable to derive statistical data that relates to substantially horizontal movements of the athlete, substantially vertical movements of the athlete, angular and/or rotational movements of the athlete, and/or energy, intensity and force expenditures by the athlete during the performance of an athletic activity. For example, in certain embodiments, the sensor assembly **216** and/or the performance sensors **216P** can provide the athlete with sensed performance characteristics that are usable to derive statistical data related to number of steps taken, total distance traveled, distance traveled per step (such as stride length), stride duration, ground contact (foot strike) duration and pattern, foot acceleration and foot angle during strides (gait tracking, including foot strike angle), speed of travel, horizontal burst (such as sudden acceleration from an average rate of speed), number of jumps, height of jumps, jump duration, vertical burst (such as take-off velocity or acceleration for a jump), number of accelerations (relating to horizontal burst and/or vertical burst), angular, twisting or rotational movements of the athlete (and/or the speed of such movements), energy expended during athletic performance (such as in kcal), and/or force expended during athletic performance (such as in psi, kpi, or other force measurements). In some embodiments, the sensor assembly **216** and/or the performance sensors **216P** can also provide the athlete with sensed performance characteristics that are usable to derive average, minimum and maximum values for any of the noted statistical data points that are generated during the performance of an athletic activity. In many embodiments, the sensor assembly **216** and/or the performance sensors **216P** can further provide the athlete with sensed

performance characteristics that are usable to derive other desired statistical data. It is appreciated that the terms “athletic performance” and “athletic activity” are used interchangeably herein.

(32) In certain implementations, such as when the user has separate feedback assemblies **15** attached to each shoe **10A**, **10B**, the feedback assemblies **15** are better able to accomplish “true” gait tracking (vs. extrapolating using only one foot). Thus, the use of separate feedback assemblies **15** for each foot enables much improved accuracy related to the various statistical data points noted above. Using separate feedback assemblies **15** on each shoe **10A**, **10B** also enables improved tracking of any gait imbalance a lot more accurately, which is linked to evaluation of proper biomechanics. For example, an injury and/or fatigue may cause an athlete to use one foot differently, leading to more joint/muscle stress. Using separate feedback assemblies **15** on each shoe **10A**, **10B** enables enhanced tracking of any such imbalances with much greater accuracy.

(33) In various implementations, the statistical data that is provided through use of the sensor assembly **216** can be subsequently utilized by the athlete to evaluate various performance metrics. The performance metrics that are assessed throughout an athletic performance can then be used by the athlete to tailor their training programs and schedules with the goal of ultimately improving their athletic performance through concepts such as improved biomechanics, injury prevention, etc.

(34) In certain embodiments or applications, the sensor assembly **216** can further include Bluetooth and/or GPS capabilities. For example, in some such embodiments, the sensor assembly **216** can further include one or more locational sensors **216L** (illustrated in FIG. 2F), such as GPS sensors, for providing accurate and precise locational information that can be used by the individual wearing the lace adjuster **14**.

(35) In some applications, the locational sensors **216L** can be utilized for purposes of navigation so that the individual wearing the lace adjuster **14** always knows where he or she is, as well as where he or she needs to go to reach any desired destination. In such uses, the locational sensors **216L** can be utilized to inhibit the person wearing the lace adjuster **14** from getting lost and/or to enable the wearer to follow a prescribed trail, such as during an adventure race and/or when exploring the wilderness.

(36) In other applications, the locational sensors **216L** can offer a sense of security for someone, such as a parent or guardian, who is charged with care for and/or monitoring of the individual wearing the lace adjuster **14**. In such applications, the locational information from the locational sensors **216L** can be wirelessly transmitted to a remote receiver so that the parent or guardian can always have the accurate and precise locational information of the person wearing the lace adjuster **14**. With such applications, the parent or guardian can help assist the wearer from getting lost and/or inhibit the wearer from going to undesired or inappropriate locations.

(37) In various embodiments, it is appreciated that any information from the sensor assembly **216**, including information from any of the performance sensors **216P** and/or the locational sensors **216L**, can be downloaded into a remote device **220** (illustrated in FIG. 2F) via a connector port **221** (illustrated in FIG. 2I), such as a USB port, or other suitable connection. As illustrated, the connector port **221** can be formed into and/or coupled to an adjuster body **22** of the lace adjuster **14**. The connector port **221** is also electrically coupled to the sensor assembly **216** and/or the image assembly **17** of the feedback assembly **15**. With such capabilities to download the desired performance characteristics from the sensor assembly **216** to the remote device **220**, the user can view any associated data that was generated during the athletic activity from any of the performance sensors **216P** and/or the locational sensors **216L** of the sensor assembly **216**. For example, the user can download information into the remote device **220** that was generated using the locational sensors **216L**, so the user can precisely see the specific path or trail that was followed, such as on foot, by bicycle, etc.

(38) The image assembly **17** can be uniquely designed and/or positioned to provide the athlete who is using the image assembly **17**, such as in conjunction with the lace adjuster **14**, with unique viewpoints from which the athlete is able to visualize and/or evaluate various aspects of their

athletic performance. For example, in different embodiments, depending upon the specific positioning and orientation of the image assembly **17** during use, the athlete is able to effectively capture, review and analyze images (such as still images and/or video images) of themselves demonstrating unique perspectives and angles of their athletic performance. With such design, the athlete may be able to gather unique insights into their athletic performance, which would not otherwise be available from remote positioning of an image assembly.

(39) For example, the image assembly **17** can provide low-resolution or high-resolution images or video (and sound). The images or video can be transmitted via Wi-Fi, Bluetooth, or a USB port in certain non-exclusive embodiments. In some embodiments, the images or video can be transmitted for a TV broadcast during a performance or game. The image assembly **17** can be controlled by a button on the lace adjuster **14** or it can be remotely controlled. In the embodiment illustrated in FIG. **1**, the image assembly **17** is secured to, mechanically coupled to and/or integrated into the lace adjuster **14**. The image, video and sound can be of the person wearing the image assembly **17** and his surrounding environment.

(40) In certain embodiments, the image assembly **17** can be directed in a generally upward or outward direction from the shoe **10A**, **10B** to capture the desired images or video. In some embodiments, the direction of where the image assembly **17** is directed can be controlled and/or adjusted by the user, and/or can be controlled remotely by another individual. Alternatively, the image assembly **17** can be directed in a different direction.

(41) Moreover, as with the sensor assembly **216**, it should be appreciated that any information from the image assembly **17** can also be downloaded into the remote device **220** via the connector port **221** or other suitable connection. With such design, the user can view any images from the athletic activity at his or her convenience after completion of the athletic activity.

(42) In certain implementations, any data and information gathered via the feedback assemblies **15**, i.e. from the sensor assemblies **216** and/or the image assemblies **17** can be connected to a module for use in a team-based aspect. More particularly, such data and information gathered via the feedback assemblies **15** can be compiled together for multiple users or athletes as part of a team evaluation or analysis within such module. In such implementations, the users can see live team-based data, which may be usable for any high performance workouts and team sports.

(43) In other implementations, any data and information gathered via the feedback assemblies **15**, i.e. from the sensor assemblies **216** and/or the image assemblies **17** can be integrated within a video game. For example, the movements of the user during an athletic performance can be utilized and/or demonstrated through corresponding movements of a video game character during the playing of the video game.

(44) In various embodiments, the lace adjuster **14** can have any suitable design for purposes of enabling the user to quickly and easily adjust, tighten or loosen the shoelace **12** of the shoe **10A**, **10B**. For example, in certain non-exclusive alternative embodiments, the lace adjuster **14** can be designed to include various features and limitations such as described in U.S. Pat. No. 8,181,320 B2 issued on May 22, 2012, and entitled "LACE ADJUSTER", U.S. Pat. No. 10,512,304 B2 issued on Dec. 24, 2019, and entitled "LACE ADJUSTER WITH INTERCHANGEABLE COVERS", and/or U.S. Pat. No. 10,595,581 B2 issued on Mar. 24, 2020, and entitled "LACE ADJUSTER ASSEMBLY INCLUDING FEEDBACK ASSEMBLY FOR USE IN VISUALIZING AND MEASURING ATHLETIC PERFORMANCE". As far as permitted, the contents of U.S. Pat. Nos. 8,181,320 B2, 10,512,304 B2, and 10,595,581 B2 are incorporated in their entireties herein by reference. Alternatively, the lace adjuster **14** can have another suitable design.

(45) As shown in the embodiment illustrated in FIG. **1**, the lace adjuster **14** can include the adjuster body **22** that is configured to be selectively coupled to the shoelace **12** of the shoe **10A**, **10B**, and a lace end retainer **23** that is coupled to the adjuster body **22**. In certain embodiments, the adjuster body **22** can include a first body member **24** and a second body member **25** that are movable relative to one another between an unlocked configuration wherein the lace adjuster **14** can

effectively receive the first lace end **18** and/or the second lace end **19** of the shoelace **12**, and a locked configuration wherein the lace adjuster **14** retains the first lace end **18** and/or the second lace end **19** so that the lace adjuster **14** is fixed in position and/or is inhibited from moving relative to the shoelace **12**. Alternatively, the lace adjuster **14** can include more components or fewer components than those specifically illustrated in FIG. **1**.

(46) In many embodiments, the lace adjuster **14** is configured to be selectively coupled to the shoelace **12** when it is desired to quickly and easily adjust, tighten and/or loosen the shoelace **12** relative to the shoe body **11**. In some embodiments, the lace end retainer **23** is configured to selectively receive and securely retain the first lace end **18**, such as at or near the first end tip **18A**, and/or the second lace end **19**, such as at or near the second end tip **19A**, to inhibit the shoelace **12** from being a potential tripping hazard for the user or wearer of the shoe **10A**, **10B**. In certain embodiments, the lace end retainer **23** is configured to receive and securely retain the first lace end **18** and/or the second lace end **19** between the lace end retainer **23** and the adjuster body **22**. More particularly, in such embodiments, the lace end retainer **23** is configured such that the first lace end **18** and/or the second lace end **19** are inhibited from being moved relative to the lace end retainer **23** and the adjuster body **22** when retained by the lace end retainer **23** by a force generated by a contact pressure of the lace end retainer **23** against a surface **22A** of the adjuster body **22**. Stated in another manner, in such embodiments, the first lace end **18** and/or the second lace end **19** are inhibited from being moved relative to the lace end retainer **23** and the adjuster body **22** when retained by the lace end retainer **23** by effectively pinching the first lace end **18** and/or the second lace end **19** between the lace end retainer **23** and the surface **22A** of the adjuster body **22**.

(47) As described herein, the lace end retainer **23** can be coupled to the adjuster body **22** in any suitable manner. For example, in one embodiment, the lace end retainer **23** is fixedly coupled to the adjuster body **22**. Alternatively, in another embodiment, the lace end retainer **23** is removably coupled to the adjuster body **22**.

(48) In some embodiments, the lace end retainer **23** can be configured such that the lace end retainer **23** extends partially around the adjuster body **22** when the lace end retainer **23** is coupled to the adjuster body **22**. Alternatively, the lace end retainer **23** can be configured such that the lace end retainer **23** extends fully around the adjuster body **22** when the lace end retainer **23** is coupled to the adjuster body **22**.

(49) The lace end retainer **23** can have any suitable design for purposes of effectively receiving and retaining the first lace end **18** and/or the second lace end **19** between the lace end retainer **23** and the adjuster body **22**. In certain embodiments, such as shown in FIG. **1**, the lace end retainer **23** can include a retainer body **23A** and a retainer aperture **23B** that extends through the retainer body **23A**.

(50) It is appreciated that the first lace end **18** and/or the second lace end **19** can be retained between the lace end retainer **23** and the adjuster body **22** in any suitable manner and can be oriented in any suitable direction. In some such embodiments, the first lace end **18** and/or the second lace end **19** can extend through the retainer aperture **23B** as the lace adjuster **14** is being initially coupled to the shoelace **12**. In one embodiment, the first lace end **18** and/or the second lace end **19** can again extend through the retainer aperture **23B** before being retained between the lace end retainer **23** and the adjuster body **22**. Alternatively, in another such embodiment, the first lace end **18** and/or the second lace end **19** can be positioned so as to extend fully under the retainer body **23A** (and not back through the retainer aperture **23B**) before being retained between the lace end retainer **23** and the adjuster body **22**. Still alternatively, in still another such embodiment, the first lace end **18** and/or the second lace end **19** can extend and be retained between the lace end retainer **23** and the adjuster body **22** before the lace end **18**, **19** extends in a generally outward direction through the retainer aperture **23B**. It is appreciated that in any of such embodiments, the lace ends **18**, **19** can extend between the lace end retainer **23** and the adjuster body **22** near or toward the top of the lace adjuster **14**, near or toward the bottom of the lace adjuster **14**, or near or toward both the top and the bottom of the lace adjuster **14**.

(51) Alternatively, in other such embodiments, the lace end retainer **23** can be positioned such that the first lace end **18** and/or the second lace end **19** do not extend through the retainer aperture **23B** as the lace adjuster **14** is being initially coupled to the shoelace **12**. In such alternative embodiments, the first lace end **18** and/or the second lace end **19** can extend through the retainer aperture **23B** before being retained between the lace end retainer **23** and the adjuster body **22**, the first lace end **18** and/or the second lace end **19** can be positioned so as to extend fully under the retainer body **23A** (and not through the retainer aperture **23B**) while being retained between the lace end retainer **23** and the adjuster body **22**, or the first lace end **18** and/or the second lace end **19** can extend between the retainer body **23A** and the adjuster body **22** before extending outwardly through the retainer aperture **23B**.

(52) Still alternatively, the lace end retainer **23** can be designed without the retainer aperture **23B**, and the first lace end **18** and/or the second lace end **19** can be positioned so as to extend at least partially, if not fully under the retainer body **23A** while being retained between the lace end retainer **23** and the adjuster body **22**.

(53) FIG. 2A is a front perspective view of the lace adjuster assembly **13** illustrated in FIG. 1, the lace adjuster assembly **13** including the lace adjuster **14**, and the feedback assembly **15**, including one or more of the sensor assembly **216** (illustrated more clearly, for example, in FIG. 2F) and the image assembly **17**, that is mechanically coupled to the lace adjuster **14**. In some embodiments, the lace adjuster assembly **13** and/or the lace adjuster **14** can be lightweight and water-resistant so that it is comfortable for the user and usable in various environments.

(54) As noted above, the design of the lace adjuster **14** can be varied for purposes of enabling the user to quickly and easily adjust, tighten or loosen the shoelace **12** (illustrated in FIG. 1) of the shoe **10A**, **10B** (illustrated in FIG. 1). In various embodiments, the lace adjuster **14** can be further configured to inhibit the shoelace **12** from becoming a potential tripping hazard for the user or wearer of the shoe **10A**, **10B**. In the embodiment illustrated in FIG. 2A, the lace adjuster **14** includes the adjuster body **22** including the first body member **24** and the second body member **25**, and the lace end retainer **23** that is coupled to the adjuster body **22**.

(55) In certain embodiments, the adjuster body **22**, such as the first body member **24** and the second body member **25**, is movable between an unlocked configuration wherein the lace adjuster **14** can effectively receive the first lace end **18** (illustrated in FIG. 1) and/or the second lace end **19** (illustrated in FIG. 1) of the shoelace **12**, and a locked configuration wherein the lace adjuster **14** retains the first lace end **18** and/or the second lace end **19** so that the lace adjuster **14** is fixed in position and/or is inhibited from moving relative to the shoelace **12**. For example, in certain embodiments, the adjuster body **22** can be configured such that the second body member **25** moves relative to the first body member **24** in a plunger-like manner as the adjuster body **22** is being moved between the unlocked configuration and the locked configuration. FIG. 2A illustrates the lace adjuster **14** in the unlocked configuration.

(56) It is appreciated that when the lace adjuster **14** is coupled to the shoelace **12**, the shoelace **12** is adjustable relative to the adjuster body **22** when the adjuster body **22** is in the unlocked configuration, and the shoelace **12** is inhibited from being adjusted relative to the adjuster body **22** when the adjuster body **22** is in the locked configuration.

(57) FIG. 2B is a front perspective view of a portion of the shoelace **12** (such as a portion of the first lace end **18** and the second lace end **19**), and the lace adjuster **14** illustrated in FIG. 2A. As shown in FIG. 2B, the lace adjuster **14** is in the locked configuration. More specifically, the second body member **25** has been moved relative to the first body member **24** such that the first lace end **18** and the second lace end **19** of the shoelace **12** can be effectively retained by the adjuster body **22**, such that movement of the shoelace **12** is inhibited relative to the adjuster body **22**.

(58) FIG. 2C is a rear perspective view of the lace adjuster **14** illustrated in FIG. 2A, the lace adjuster **14** again being shown in the unlocked configuration; and FIG. 2D is a rear perspective view of the lace adjuster **14** illustrated in FIG. 2A, the lace adjuster **14** again being shown in the

locked configuration.

(59) Looking at FIGS. 2A-2D together, the first body member **24** includes one or more front apertures **227** (two are illustrated, for example, in FIG. 2A) and one or more rear apertures **228** (two are illustrated, for example, in FIG. 2C), and the second body member **25** includes second apertures **229** (two are illustrated more clearly, for example, in FIG. 2B).

(60) When the lace adjuster **14** is in the process of being coupled to the shoelace **12**, the adjuster body **22** and/or the body members **24**, **25** are positioned in the unlocked configuration. When in the unlocked configuration, as shown in FIGS. 2A and 2C, the front apertures **227** (illustrated in FIG. 2A) and the rear apertures **228** (illustrated in FIG. 2C) of the first body member **24** are substantially aligned with and concentric with the second apertures **229** (illustrated in FIG. 2B) of the second body member **25**. With such design, the first lace end **18** can be positioned to extend through one of the front apertures **227** of the first body member **24**, through one of the second apertures **229** of the second body member **25**, and through one of the rear apertures **228** of the first body member **24**. Similarly, the second lace end **19** can also be positioned to extend through one of the front apertures **227** of the first body member **24**, through one of the second apertures **229** of the second body member **25**, and through one of the rear apertures **228** of the first body member **24**.

(61) Subsequently, when in the locked configuration, as shown in FIGS. 2B and 2D, the second body member **25** extends somewhat away from the first body member **24**, and the front apertures **227** (illustrated in FIG. 2A) and the rear apertures **228** (illustrated in FIG. 2C) of the first body member **24** are not substantially aligned with or concentric with the second apertures **229** (illustrated in FIG. 2B) of the second body member **25**. Thus, when in the locked configuration, the first lace end **18** (illustrated in FIG. 2B) and the second lace end **19** (illustrated in FIG. 2B) of the shoelace **12** (illustrated in FIG. 2B) can be effectively retained within the lace adjuster **14**, such that the first lace end **18** and the second lace end **19** of the shoelace **12** are inhibited from moving relative to the adjuster body **22**.

(62) As shown in this embodiment, it is appreciated that the second body member **25** fits partly within and moves up and down (such as when the lace adjuster **14** is oriented vertically) relative to the first body member **24** in a plunger-like manner as the adjuster body **22** is moved between the locked configuration and the unlocked configuration. Stated in another manner, in such embodiment, the first body member **24** is open along a top and into an upper portion **224U** (illustrated in FIG. 2B) of the first body member **24** and, as such, is designed to receive at least a portion of the second body member **25** within such open upper portion **224U** and to allow the second body member **25** to move up and down over a movement range relative to and/or at least partially within the first body member **24**, such that the first body member **24** and the second body member **25** and/or the adjuster body **22** as a whole can move between the locked configuration and the unlocked configuration.

(63) It should be appreciated that the shape of the front apertures **227**, the rear apertures **228**, and the second apertures **229** can be varied as desired. For example, in some embodiments, the front apertures **227**, the rear apertures **228**, and/or the second apertures **229** can include one or more tooth-shaped projections **230** (illustrated, for example, in FIG. 2F) that can be utilized to more effectively retain the shoelace **12** when the lace adjuster **14** is in the locked configuration. Alternatively, the front apertures **227**, the rear apertures **228**, and/or the second apertures **229** can have another suitable design.

(64) As noted above, the lace end retainer **23** is coupled to the adjuster body **22**. In some embodiments, the lace end retainer **23** is specifically configured to inhibit the first lace end **18** and/or the second lace end **19** from being a potential tripping hazard by inhibiting the first lace end **18** and/or the second lace end **19** from being moved relative to the lace end retainer **23** and the adjuster body **22** when retained by the lace end retainer **23** by a force generated by a contact pressure of the lace end retainer **23** against a surface **22A** of the adjuster body **22**.

(65) The lace end retainer **23** can have any suitable design and can be coupled to the adjuster body

22 in any suitable manner. For example, in certain embodiments, as shown in FIGS. 2A and 2B, the lace end retainer 23 can include the retainer body 23A and the retainer aperture 23B that extends through the retainer body 23A. As illustrated, the lace end retainer 23 can be positioned such that the retainer aperture 23B is substantially aligned with the front apertures 227 and the rear apertures 228 formed in the first body member 24 of the adjuster body 22 (and also substantially aligned with the second apertures 229 of the second body member 25 when the adjuster body 22 is in the unlocked configuration). With such design, when the first lace end 18 and/or the second lace end 19 of the shoelace 12 are positioned to extend through the front apertures 227, the second apertures 229 and the rear apertures 228, the lace ends 18, 19 could also easily extend through the retainer aperture 23B. Moreover, when it is desired to effectively retain the first lace end 18 and/or the second lace end 19 with the lace end retainer 23, the first lace end 18 and/or the second lace end 19 can be positioned back through the retainer aperture 23B before being retained, such as pinched, between the retainer body 23A and the surface 22A of the adjuster body 22.

(66) As shown in this embodiment, the lace end retainer 23 can further include at least one first coupling member 223C, such as a coupling aperture that extends through retainer body 23A, with each of the at least one first coupling member 223C being configured to engage a second coupling member 231 of the adjuster body 22, such as a coupling projection that extends away from the adjuster body 22. In one embodiment, the lace end retainer 23 can be configured to extend partially about the adjuster body 22 when the lace end retainer 23 is coupled to the adjuster body 22. In such embodiment, the retainer body 23A can include two first coupling members 223C (such as two coupling apertures), with each of the two first coupling members 223C being positioned about a different second coupling member 231 that extends or projects away from the adjuster body 22. Alternatively, in another such embodiment, the lace end retainer 23 can be configured to extend fully about the adjuster body 22 when the lace end retainer 23 is coupled to the adjuster body 22. In such alternative embodiment, the retainer body 23A can include two first coupling members 223C (such as two coupling apertures), with each of the two first coupling members 223C being positioned about a single second coupling member 231 that extends or projects away from the adjuster body 22. Still alternatively, in still another such embodiment, the lace end retainer 23 can be configured to extend fully about the adjuster body 22 with a general loop-type design, such that the retainer body 23A does not need any coupling apertures and there are no coupling members that extend or project away from the adjuster body 22. Yet alternatively, the lace end retainer 23 can be coupled to the adjuster body 22 in another suitable manner.

(67) The lace end retainer 23 can be formed from any suitable materials. For example, in some embodiments, the lace end retainer 23 is formed from a resilient material, such as rubber or another suitable elastic or resilient material. In such embodiments, the lace end retainer 23 can be stretched at least slightly when the lace end retainer 23 is coupled to the adjuster body 22. With such design, based on the resilient nature of the lace end retainer 23, the lace end retainer 23 is better able to exert a force onto the surface 22A of the adjuster body 22 based on a contact pressure between the lace end retainer 23 and the surface 22A of the adjuster body 22. Thus, the lace end retainer 23 is better able to pinch the first lace end 18 and/or the second lace end 19 between the lace end retainer 23 and the surface 22A of the adjuster body 22. Alternatively, the lace end retainer 23 can be formed from another suitable material.

(68) Referring now to FIGS. 2C and 2D, as shown, the lace adjuster 14 can further include a motion restrictor 232 that is coupled to and cantilevers away from the adjuster body 22. In particular, in this embodiment, the motion restrictor 232 includes a first restrictor end 232A that is secured to the first body member 24 and a second restrictor end 232B that is spaced apart from and/or is not directly secured to the first body member 24. With the design illustrated in the Figures, the motion restrictor 232 is designed similar to a spring-type clip, which is configured to extend underneath at least a portion of the shoelace 12 (illustrated in FIG. 2B) so that at least a portion of the motion restrictor 232 is positioned substantially directly between the shoelace 12 and

the shoe body **11** (illustrated in FIG. 1) of the shoe **10A**, **10B** (illustrated in FIG. 1). Thus, the lace adjuster **14** is inhibited from bouncing around and is held more firmly in position when coupled to the shoelace **12** and when the user is engaging in various types of activities.

(69) FIG. 2E is a top view of the lace adjuster **14** illustrated in FIG. 2A; FIG. 2F is a cutaway view of the lace adjuster **14** taken on line F-F in FIG. 2E, the lace adjuster **14** being shown in the unlocked configuration; and FIG. 2G is a comparable sectional view of the lace adjuster **14** illustrated in FIG. 2A, the lace adjuster **14** being shown in the locked configuration. As shown, FIGS. 2E-2G illustrate certain additional features or components that can be included in certain embodiments of the lace adjuster **14**.

(70) For example, FIGS. 2F and 2G illustrate certain additional aspects of the movement of the lace adjuster **14**, such as the relative movement of the first body member **24** and the second body member **25** of the adjuster body **22**, between the unlocked configuration and the locked configuration. More specifically, as illustrated, the first body member **24** and the second body member **25** are resiliently coupled to one another with one or more resilient members **233** (only one is illustrated in this example). In particular, the resilient member **233** is connected to and extends between the first body member **24** and the second body member **25** to enable the adjuster body **22** to resiliently move between the unlocked configuration and the locked configuration. It is appreciated that the resilient member **233** can be connected to each of the first body member **24** and the second body member **25** in any suitable manner. For example, in one non-exclusive embodiment, each of the first body member **24** and the second body member **25** can include a member receiver (not shown) that is adapted to receive and retain a portion of the resilient member **233** in order to secure the resilient member **233** to the first body member **24** and the second body member **25**, respectively. Alternatively, the resilient member **233** can be connected to the first body member **24** and/or the second body member **25** in another suitable manner.

(71) The design of the resilient member **233** can be varied depending on the requirements of the lace adjuster **14**. For example, in the embodiment illustrated in FIGS. 2F and 2G, the resilient member **233** is a spring. In one embodiment, the resilient member **233** is a stiff spring that can hold the first body member **24** and the second body member **25** substantially straight relative to one another to ease the movement of the body members **24**, **25** between the locked configuration and the unlocked configuration. Alternatively, the resilient member **233** can be another piece of resilient material.

(72) In this embodiment, the resilient member **233** urges the second body member **25** up and/or away relative to the first body member **24** so that the adjuster body **22** is urged and/or biased toward the locked configuration. Alternatively, the resilient member **233** can be designed to urge the second body member **25** within the first body member **24** so that the adjuster body **22** is urged and/or biased toward the unlocked configuration. In such alternative embodiment, the lace adjuster **14** would further require a locking mechanism (not illustrated) that would maintain the first body member **24** and the second body member **25** in the locked configuration. In these alternative embodiments, the resilient member **233** is either extended or compressed as the first body member **24** and the second body member **25** are moved between the locked configuration and the unlocked configuration. Still alternatively, in one embodiment, the lace adjuster **14** can further include a stop (not shown) that inhibits and/or stops relative movement between the body members **24**, **25** so that the body members **24**, **25** are inhibited from moving beyond the desired positioning for the body members **24**, **25** when in the locked configuration and the unlocked configuration. It is appreciated that in such embodiments the stop can be positioned in different manners depending on in which direction the resilient member **233** is biased.

(73) In certain embodiments, the lace adjuster **14** can further include a guide system (not shown) that guides relative movement between the first body member **24** and the second body member **25** as the adjuster body **22** is moved between the unlocked configuration and the locked configuration. In such embodiments, the guide system can have any suitable design that enables controlled

relative movement between the first body member **24** and the second body member **25** as the adjuster body **22** is moved between the unlocked configuration and the locked configuration. Alternatively, in other embodiments, the lace adjuster **14** can be designed without a specific guide system. In some such alternative embodiments, as noted above, the relative movement between the body members **24**, **25** can be guided through use of the stiff spring as the resilient member **233**. (74) As shown in FIGS. 2F and 2G, the lace adjuster **14** can further include the feedback assembly **15**, including the sensor assembly **216** and/or the image assembly **17**, that is mechanically coupled to the adjuster body **22** and/or another portion of the lace adjuster **14**. In particular, in this embodiment, the feedback assembly **15** is positioned substantially within a body cavity **234** that is formed within the adjuster body **22**. In some embodiments, the body cavity **234** can be provided in the form of a sealed and/or water-resistant chamber that can be utilized to provide greater protection from the surrounding environment for the feedback assembly **15**. Alternatively, the feedback assembly **15** can be mechanically coupled to another portion of the lace adjuster **14** and/or the adjuster body **22**. For example, in certain non-exclusive alternative embodiments, the lace adjuster **14** can include an adjuster cover plate **238** (illustrated in FIG. 2J) that is coupled to the adjuster body **22**, and at least a portion of the feedback assembly **15** can be mechanically coupled to the adjuster cover plate **238**.

(75) In certain embodiments, the body cavity **234** can be formed, at least in part, within and/or adjacent to the first body member **24**. Alternatively, in other embodiments, the body cavity **234** can be formed, at least in part, between the first body member **24** and the second body member **25** of the adjuster body **22**. Still alternatively, the feedback assembly **15** can be positioned on, coupled to and/or incorporated within the lace adjuster **14** in another suitable manner.

(76) In some embodiments, the adjuster body **22** can further include a separator **235**, such as a separation wall, that can be used to isolate the body cavity **234**, within which the feedback assembly **15** is retained, from the open upper portion **224U** (illustrated in FIG. 2B) of the first body member **24**, within which the second body member **25** moves during movement of the adjuster body **22** between the unlocked configuration and the locked configuration. With such design, the feedback assembly **15** can be better protected from the surrounding environment.

(77) As described herein, the feedback assembly **15** is usable by the user to provide statistical data and/or images of the user, such as during an athletic performance, in order to effectively gauge various aspects of their athletic performance. In some embodiments, the feedback assembly **15** can be substantially similar in design and function to the feedback assemblies that are incorporated within the lace adjusters illustrated and described in U.S. Pat. No. 10,595,581 B2 issued on Mar. 24, 2020, and entitled “LACE ADJUSTER ASSEMBLY INCLUDING FEEDBACK ASSEMBLY FOR USE IN VISUALIZING AND MEASURING ATHLETIC PERFORMANCE”, and/or in U.S. application Ser. No. 17/354,655 filed on Jun. 22, 2021, and entitled “LACE ADJUSTER”, which have, to the extent allowable, been incorporated herein in their entirety.

(78) In some embodiments, the sensor assembly **216** can be uniquely designed to incorporate one or more performance sensors **216P** that are configured to sense various performance characteristics for the user during an athletic performance. The sensed performance characteristics can be subsequently utilized to provide the athlete or user who is using the sensor assembly **216**, such as in conjunction with the lace adjuster **14**, with statistical data and/or performance measurables that enable the athlete to effectively gauge various aspects of their athletic performance. For example, in certain embodiments, the one or more performance sensors **216P** can include one or more two-axis accelerometers, a three-axis accelerometer, a three-axis gyrometer (or gyroscope) and/or another type of rate sensor, and/or a three-axis magnetometer. Additionally, and/or alternatively, the one or more performance sensors **216P** can include additional appropriate sensor types. Further, in some embodiments, the sensor assembly **216** can also include a real-time clock **216R** that enables more accurate time tracking.

(79) In different embodiments, as noted above, the sensor assembly **216** can include the one or

more performance sensors **216P** in order to provide sensed performance characteristics that are usable to derive statistical data that relates to substantially horizontal movements of the athlete, substantially vertical movements of the athlete, angular and/or rotational movements of the athlete, and/or energy, intensity and force expenditures by the athlete during the performance of an athletic activity. For example, in certain embodiments, the sensor assembly **216** and/or the performance sensors **216P** can provide the athlete with sensed performance characteristics that are usable to derive statistical data related to number of steps taken, total distance traveled, distance traveled per step (such as stride length), stride duration, ground contact duration and pattern, foot acceleration and foot angle during strides (gait tracking), speed of travel, horizontal burst (such as sudden acceleration from an average rate of speed), number of jumps, height of jumps, jump duration, vertical burst (such as take-off velocity or acceleration for a jump), number of accelerations (relating to horizontal burst and/or vertical burst), angular, twisting or rotational movements of the athlete (and/or the speed of such movements), energy expended during athletic performance (such as in kcal), and/or force expended during athletic performance (such as in psi, kpi, or other force measurements). In some embodiments, the sensor assembly **216** and/or the performance sensors **216P** can also provide the athlete with sensed performance characteristics that are usable to derive average, minimum and maximum values for any of the noted statistical data points that are generated during the performance of an athletic activity. In many embodiments, the sensor assembly **216** and/or the performance sensors **216P** can further provide the athlete with sensed performance characteristics that are usable to derive other desired statistical data. In some embodiments, the statistical data that is provided by the sensor assembly **216** can be subsequently utilized by the athlete to tailor their training programs and schedules with the goal of ultimately improving their athletic performance. Moreover, the athlete can further compare the statistical data gathered during different and/or subsequent athletic performances to better evaluate any changes of performance measurables.

(80) It is appreciated that any and all of the performance characteristics measured and/or sensed by the one or more performance sensors **216P** can be combined in any suitable manner to enable the generation of various statistical data and/or performance measurables for the athlete during the performance of an athletic activity or event. It is further appreciated that in order to more effectively evaluate the various statistical data from the athletic performances, the athlete may desire to provide certain input information, such as the height and weight of the athlete. In one embodiment, the athlete may manually input such information as height and weight into the sensor assembly **216** via a remote device **220** (illustrated as a box that is not to scale), such as a smartphone, a smart watch, a tablet, a computer, and/or any other suitable computing device. Alternatively, information such as the height and weight of the athlete can be provided to the sensor assembly **216** in another suitable manner. This information can further be utilized to see the effects of people's height and weight on the performance data. It is also appreciated that any statistical data related to energy expended and/or force expended can require information such as the weight of the athlete in order for such statistical data to be accurately generated.

(81) In certain embodiments or applications, the sensor assembly **216** can additionally and/or alternatively include the one or more locational sensors **216L**, such as GPS sensors, for providing accurate and precise locational information that can be used by the individual wearing the lace adjuster **14**. For example, in certain non-exclusive alternative applications, the locational sensors **216L** can be utilized for purposes of navigation and/or the locational sensors **216L** can be utilized for purposes of tracking movements of the user. With such applications, the individual wearing the lace adjuster **14** always knows where he or she is, as well as where he or she needs to go to reach any desired destination. In such uses, the locational sensors **216L** can be utilized to inhibit the person wearing the lace adjuster **14** from getting lost and/or to enable the wearer to follow a prescribed trail, such as during an adventure race or when exploring the wilderness. Moreover, the locational sensors **216L** can offer a sense of security for someone, such as a parent or guardian,

who is charged with care for and/or monitoring of the individual wearing the lace adjuster **14**. In such applications, the locational information from the locational sensors **216L** can be wirelessly transmitted to the remote device **220** so that the user and/or the parent or guardian can always have the accurate and precise locational information of the person wearing the lace adjuster **14**. The locational sensors **216L** can also be used to track the movement of the user. For example, the route ran or biked can be recorded and stored for future analysis. Other information, such as time and altitude can also be recorded and stored for future analysis.

(82) Moreover, as described herein below, in some applications, the locational sensors **216L** can be utilized in conjunction with additional locational sensors, such as GPS sensors, or beacons that are positioned remotely from the lace adjuster **14**, such as being positioned on or near an athletic field or court, to more precisely and accurately provide locational information for the user.

(83) The image assembly **17**, such as a digital camera in some embodiments, can be configured and/or positioned to provide the user with unique viewpoints from which the user is able to visualize and/or evaluate various aspects of their athletic performance. For example, in different embodiments, depending upon the specific positioning and orientation of the image assembly **17** during use, the user is able to effectively capture, review and analyze images (such as still images and/or video images) of themselves demonstrating unique perspectives and angles of their athletic performance. For example, the image assembly **17** can provide low-resolution or high-resolution images or video (and sound). The images or video can be transmitted via Wi-Fi, Bluetooth, or a USB port. In certain embodiments, the images or video can be transmitted for a TV broadcast during a performance or game. The image assembly **17** can be controlled by a button on the lace adjuster **14** or it can be remotely controlled.

(84) In certain embodiments, the image assembly **17** can be directed in a generally upward or outward direction from the shoe **10A**, **10B** (illustrated in FIG. **1**) to capture the desired images or video. With such design, the user may be able to gather unique insights into their athletic performance, which would not otherwise be available from remote positioning of an image capturing assembly. Alternatively, the image assembly **17** can be directed in a different direction. In certain embodiments, the direction of where the image assembly **17** is directed can be controlled and/or adjusted by the user, and/or can be controlled remotely by another individual.

(85) In some embodiments, the adjuster body **22** can include an imaging aperture **236** (illustrated in FIG. **2A**) through which the image assembly **17** is able to capture images of the user during use.

(86) It is appreciated that through use of the motion restrictor **232** (illustrated in FIG. **2C**), which inhibits the lace adjuster **14** from bouncing around during use, the feedback assembly **15** is able to provide more precise, accurate and clear sensed information from the sensor assembly **216** and images from the image assembly **17**.

(87) Moreover, it is appreciated that any information from the feedback assembly **15**, such as from the sensor assembly **216** and/or the image assembly **17**, can be downloaded or transmitted into the remote device **220** in any suitable manner. For example, in certain embodiments, the information from the feedback assembly **15**, such as from the sensor assembly **216** and/or the image assembly **17**, can be downloaded or transmitted into the remote device **220** via Bluetooth, Wi-Fi, or another suitable connection. It is further appreciated that any such information from the feedback assembly **15** can be downloaded or transmitted to the remote device **220** wirelessly or via a wired connection. Certain specific embodiments of the feedback assembly **15**, including the sensor assembly **216** and/or the image assembly **17**, and the remote device **220** will be described in greater detail herein below in relation to FIGS. **3** and **4**.

(88) In some embodiments, a power source **215E** (illustrated as a box in phantom) can be included to provide necessary power for both the sensor assembly **216** and the image assembly **17** of the feedback assembly **15**; or a separate power source can be included for each of the sensor assembly **216** and the image assembly **17** of the feedback assembly **15**.

(89) The power source **215E** can have any suitable design for purposes of providing necessary

power for both the sensor assembly **216** and the image assembly **17** of the feedback assembly **15**. For example, in some embodiments, the power source **215E** can include one or more batteries. In a specific example, the one or more batteries can be selectively recharged via the connector port **221** (illustrated in FIG. **2I**). Additionally, or in the alternative, the connector port **221** can be used for other suitable purposes. For example, in some alternative embodiments, the connector port **221** can also be utilized for purposes of transmitting information from the lace adjuster **14** to the remote device **220**. Still alternatively, in some embodiments, the power source **215E** can be charged remotely.

(90) FIG. **2H** is a front perspective view of the lace adjuster **14** illustrated in FIG. **2A**.

(91) FIG. **2I** is another front perspective view of the lace adjuster **14** illustrated in FIG. **2A**. As shown in FIG. **2I**, the lace adjuster **14** includes a port cover **237** that is coupled to the adjuster body **22**, and that can be selectively opened to reveal the connector port **221** that is usable for charging the power source **215E** (illustrated, for example, in FIG. **2F**), and/or for transmitting information from the feedback assembly **15** (illustrated, for example, in FIG. **2F**) to the remote device **220** (illustrated, for example, in FIG. **2F**), such as a smartphone, a smart watch, a tablet, a computer, and/or any other suitable computing device.

(92) FIG. **2J** is still another front perspective view of the lace adjuster **14** illustrated in FIG. **2A**. As shown, FIG. **2J** illustrates one or more additional features of the lace adjuster **14**, such as the selective coupling of an adjuster cover plate **238** to the first body member **24** so as to form a portion of the adjuster body **22**. In some embodiments, the adjuster cover plate **238** can include a design (not shown) so as to give the lace adjuster **14** a more interesting appearance. In various embodiments, the adjuster cover plate **238** can be interchangeable with other alternative adjuster cover plates so that the lace adjuster **14** can have any desired design as included within the adjuster cover plate **238**.

(93) It is appreciated that the adjuster cover plate **238** can be selectively attached to and detached from the first body member **24** and/or the adjuster body **22** in any suitable manner. For example, in certain embodiments, the adjuster cover plate **238** can include a first attachment member **239** that is configured to selectively engage a second attachment member **240** that is coupled to and/or included within the first body member **24** or another portion of the adjuster body **22**. In one such embodiment, the first attachment member **239** can include a hook-type element that is configured to engage a groove-type element of the second attachment member **240**. Alternatively, the first attachment member **239** can include a groove-type member that is configured to be engaged by a hook-type member of the second attachment member **240**. Still alternatively, the first attachment member **239** and/or the second attachment member **240** can have another suitable design.

(94) It is appreciated that in various embodiments, the adjuster cover plate **238** can include two first attachment members **239** and the first body member **24** (or other portion of the adjuster body **22**) can include two second attachment members **240** so that the adjuster cover plate **238** is selectively attachable to the first body member **24** and/or the adjuster body **22** on two spaced apart locations, such as on opposing sides of the adjuster body **22**. It is further appreciated that the adjuster cover plate **238** can be attached to the adjuster body **22**, such as to the first body member **24**, at more than one place on each side of the adjuster body **22**. For example, in one such alternative embodiment, the adjuster cover plate **238** can be attached to the first body member **24** at one place on one side of the adjuster body **22**, and the adjuster cover plate **238** can be attached to the first body member **24** at two spaced apart places on the other side of the adjuster body **22**.

(95) Yet alternatively, the adjuster cover plate **238** can be hingedly coupled to the adjuster body **22** on one side of the adjuster cover plate **238**. With such design, the adjuster cover plate **238** can be moved relative to the adjuster body **22**, such as similar to the opening of a door, to provide access to the feedback assembly **15** (illustrated in FIG. **2F**) that is positioned substantially within the body cavity **234** (illustrated in FIG. **2F**), which in certain embodiments can be defined between the first body member **24** and the adjuster cover plate **238**. Moreover, in certain embodiments, the feedback

assembly **15** can be coupled to, or positioned on and/or substantially adjacent to the adjuster cover plate **238**.

(96) Additionally, or in the alternative, the lace adjuster **14** that includes such interchangeable adjuster cover plates **238** can be designed such as is illustrated and described in U.S. Pat. No. 10,512,304 B2 issued on Dec. 24, 2019, and entitled “LACE ADJUSTER WITH INTERCHANGEABLE COVERS”, which has, to the extent allowable, been incorporated herein in its entirety.

(97) Moreover, in certain embodiments, the lace adjuster **14** can further include a light assembly (not shown) including one or more lights (not shown), such as LED lights, that can be mounted on and/or positioned substantially adjacent to the adjuster cover plate **238** or another component of the lace adjuster **14**. In particular, in some such embodiments, the lights can be coupled to the adjuster cover plate **238** and/or can be positioned such that the lights can shine and/or extend through one or more light apertures (not shown) in the adjuster cover plate **238**. Such lights can also be positioned so as to more effectively and dramatically draw attention to the design on the adjuster cover plate **238** and/or to provide desired lighting for someone using the lace adjuster **14** in less favorable lighting situations, such as at night. Additionally, and/or alternatively, the light assembly and/or the lights can be positioned in a different area of the lace adjuster **14**.

(98) FIGS. 2K-2R illustrate certain additional views of the lace adjuster **14** illustrated in FIG. 2A, and thus provide different vantage points of various features and components of the lace adjuster **14**. In particular, FIG. 2K is a rear perspective view of the lace adjuster **14** illustrated in FIG. 2A; FIG. 2L is another rear perspective view of the lace adjuster **14** illustrated in FIG. 2A; FIG. 2M is a bottom view of the lace adjuster **14** illustrated in FIG. 2A; FIG. 2N is another bottom view of the lace adjuster **14** illustrated in FIG. 2A; FIG. 2O is a side view of the lace adjuster **14** illustrated in FIG. 2A; FIG. 2P is a front view of the lace adjuster **14** illustrated in FIG. 2A; FIG. 2Q is another side view of the lace adjuster **14** illustrated in FIG. 2A; and FIG. 2R is a rear view of the lace adjuster **14** illustrated in FIG. 2A.

(99) FIG. 3 is a simplified schematic illustration of an embodiment of the sensor assembly **316**. FIG. 3 further includes a simplified schematic illustration of an embodiment of the remote device **320** that can be utilized in conjunction with and/or as part of the sensor assembly **316**. This sensor assembly **316** can be used in each of the lace adjuster assemblies **13** illustrated in FIG. 1.

(100) The design of the sensor assembly **316** can be varied. For example, as illustrated in FIG. 3, the sensor assembly **316** can include an assembly body **350**, one or more performance sensors **316P** (four are illustrated as boxes in phantom in FIG. 3), one or more locational sensors **316L** (illustrated as a box in phantom), a real-time clock **352** (illustrated as a box in phantom), a storage device **354** (illustrated as a box in phantom), a transmitter **356** (illustrated as a box in phantom), a receiver **358** (illustrated as a box in phantom), a controller **360** (illustrated as a box in phantom), and a power source **362** (illustrated in phantom). As shown, in this embodiment, each of the one or more performance sensors **316P**, the one or more locational sensors **316L**, the real-time clock **352**, the storage device **354**, the transmitter **356**, the receiver **358**, the controller **360**, and the power source **362** can be coupled to and/or positioned substantially within the assembly body **350**. The design of each of these components can be varied to suit the design requirements of the sensor assembly **316**. Alternatively, the sensor assembly **316** can have another suitable design, which can comprise more or fewer components than those specifically illustrated in FIG. 3. Still alternatively, one or more of the components can be provided remotely from the assembly body **350**.

(101) As shown, in one embodiment, the assembly body **350** can provide a housing for the one or more performance sensors **316P**, the one or more locational sensors **316L**, the real-time clock **352**, the storage device **354**, the transmitter **356**, the receiver **358**, the controller **360**, and the power source **362**. The design of the assembly body **350** can be varied. For example, in one embodiment, the assembly body **350** is substantially rectangular box-shaped. Alternatively, the assembly body **350** can have another suitable shape.

(102) As noted above, the sensor assembly **316** can provide the athlete with various statistical data and/or performance measurables that enable the athlete to effectively gauge various aspects of their athletic performance. In order to effectively provide such statistical data and/or performance measurables, the sensor assembly **316** needs the one or more performance sensors **316P** and the one or more locational sensors **316L** to encompass certain features in order to sense the appropriate performance variables. For example, in certain embodiments, the one or more performance sensors **316P** can include one or more two-axis accelerometers, a three-axis accelerometer, a three-axis gyrometer (or gyroscope) and/or another type of rate sensor, and/or a three-axis magnetometer. In some embodiments, the one or more performance sensors **316P** can include additional appropriate sensor types.

(103) As discussed herein, the one or more performance sensors **316P** can be effectively utilized to sense various performance characteristics, which can be subsequently utilized to derive and/or generate usable statistical data and/or performance measurables for the athlete. For example, the two-axis accelerometers can be utilized to measure and/or sense acceleration of the athlete during his or her performance along two axes. More specifically, one two-axis accelerometer can be utilized to measure and/or sense acceleration of the athlete along the horizontal axes (such as the X axis and the Y axis); and other two-axis accelerometers can be utilized to measure and/or sense acceleration of the athlete along one horizontal axis (such as either the X axis or the Y axis) and the vertical axis (such as the Z axis). The three-axis accelerometer can be utilized to measure and/or sense acceleration of the athlete along all three axes (such as along the X axis, the Y axis, and the Z axis). It should be appreciated that by comparing the performance characteristics measured and/or sensed by the three-axis accelerometers to the performance characteristics measured and/or sensed by each of the two-axis accelerometers (such as by subtracting two-axis data from the three-axis data), accurate acceleration data can be determined along each individual axis to effectively isolate vertical and horizontal acceleration of the athlete.

(104) The three-axis gyrometer (or gyroscope) or other type of rate sensor can be utilized to measure and/or sense orientation information for the athlete in three dimensions (such as about the X axis, the Y axis, and the Z axis) as a means to ultimately provide usable data with regard to angular movements of the athlete (such as twist and rotation) during performance of the athletic activity or event.

(105) The three-axis magnetometer can be utilized to sense and/or track the Earth's magnetic field, and thus to measure the strength (such as magnitude) and direction of magnetic fields at a point in space in relation to the various movements of the athlete.

(106) Using foot acceleration measurements, the sensor assembly **316** can then estimate the foot position in space relative to the ground. The sensor assembly **316** can then extract metrics such as stride duration, stride length, ground contact (foot strike) duration, foot angles (at all times through strides, including foot strike angles), etc., which are what coaches need to track performance. For example, a sprinter will work on spending as little time on the ground and increasing stride duration and/or stride length. Recording those metrics continuously during a session/game, the athlete can then visualize how fatigue, equipment, terrain, etc., impact their performance.

(107) Another simple example is static vertical jump which is a good test and/or metric for many sports. While the sensor assembly cannot necessarily measure height directly, it can detect takeoff and landing and so it can measure the jump duration, which is directly related to height. An athlete can then test and/or track their jump performance over time.

(108) Thus, at the most basic level, the sensor assembly **316** and/or the performance sensors **316P** can be utilized to track acceleration and rotation in three dimensions. This allows the sensor assembly **316** to track the path of the foot through space, including its attitude and/or how tilted the foot is in relation to the ground (so called pitch/roll/yaw angles). In some embodiments, besides being able to count steps and cadence, the sensor assembly **316** is also usable to easily detect if the user is walking, or running, or jumping, or pivoting and record the sequence of such events (which

can be used in different sports to track game play, for example). As noted above, when applied to human foot motion, the sensor assembly **316** can also measure things like foot strike pattern, stride duration (when running), ground contact duration (when running), static jump height, etc. In theory, more sport specialized metrics are also feasible.

(109) It is appreciated that any and all of the performance characteristics measured and/or sensed by the one or more performance sensors **316P** can be combined in any suitable manner to enable the generation of various statistical data and/or performance measurables for the athlete during the performance of an athletic activity or event.

(110) Importantly, as noted above, the use of a separate sensor assembly **316** coupled to each shoe **10A**, **10B** (illustrated in FIG. 1) or each foot of the user can further enhance the accuracy and extent of the statistical data that can be derived from the performance characteristics that are sensed by the performance sensors **316P**. For example, using two separate devices (one per foot) allows for “true” gait tracking (vs. extrapolating using only one foot). Using two devices will also enable the user to track any potential gait imbalance a lot more accurately, which is linked to biomechanics and/or injury prevention. For example, an injury and/or fatigue may cause an athlete to use one foot differently, leading to more joint/muscle stress. Thus, using two separate devices enables the user to track such imbalances with more accuracy.

(111) It is further appreciated that in order to more effectively evaluate the various statistical data from the athletic performances, the athlete may desire to provide certain input information, such as the height and weight of the athlete. In one embodiment, the athlete may manually input such information as height and weight into the sensor assembly **316** and/or the controller **360**, such as via communication with the remote device **320**, such as a smartphone, a smart watch, a tablet, a computer, and/or any other suitable computing device. Alternatively, information such as the height and weight of the athlete can be provided to the sensor assembly **316** and/or the controller **360** in another suitable manner. This information can further be utilized to see the effects of people's height and weight on the performance data. It is also appreciated that any statistical data related to energy expended and/or force expended can require information such as the weight of the athlete in order for such statistical data to be accurately generated.

(112) Moreover, the athlete can further provide such information as most recent food and/or liquid intake, latest sleeping experience, most recent exercise and extent thereof, etc. as a means to help define when the athlete may be able to experience optimum performance.

(113) In certain embodiments or applications, the sensor assembly **316** can additionally and/or alternatively include the one or more locational sensors **316L**, such as GPS sensors, for providing accurate and precise locational information that can be used by the individual wearing the lace adjuster **14** (illustrated in FIG. 1). For example, in certain non-exclusive alternative applications, the locational sensors **316L** can be utilized for purposes of navigation and/or the locational sensors **316L** can be utilized for purposes of tracking movements of the user. With such applications, the individual wearing the lace adjuster **14** always knows where he or she is, as well as where he or she needs to go to reach any desired destination. In such uses, the locational sensors **316L** can be utilized to inhibit the person wearing the lace adjuster **14** from getting lost and/or to enable the wearer to follow a prescribed trail, such as during an adventure race or when exploring the wilderness. Moreover, as also noted above, the locational sensors **316L** can offer a sense of security for someone, such as a parent or guardian, who is charged with care for and/or monitoring of the individual wearing the lace adjuster **14**. In such applications, the locational information from the locational sensors **316L** can be wirelessly transmitted to the remote device **320** so that the user and/or the parent or guardian can always have the accurate and precise locational information of the person wearing the lace adjuster **14**.

(114) In many embodiments, the locational sensors **316L** can be used to track the movement of the user. For example, the route ran or biked can be recorded and stored for future analysis.

(115) Other information, such as time and altitude can also be recorded and stored for future

analysis. The real-time clock **352** enables any use of time to be tracked much more precisely than a comparable system that does not include a real-time clock. Stated in another manner, the real-time clock **352** enables more accurate time tracking as opposed to systems without a real-time clock, which can be prone to drift. The real-time clock **352** also obviates the need to rely on any timing mechanism that may be present in the remote device **320**.

(116) The data that is sensed by the one or more performance sensors **316P** and the one or more locational sensors **316L**, as well as the data input by the athlete such as via the remote device **320** (or otherwise), can be stored and/or maintained within the storage device **354** of the sensor assembly **316**. The storage device **354** can have any suitable design that enables the storing and/or maintenance of information.

(117) The transmitter **356** can be utilized to transmit the information and data that is stored within the storage device **354** (or data from the sensors **316P**, **316L**) to the controller **360** and/or the remote device **320**, such as a remote smart phone, computer, etc. The transmitter **356** can have any suitable design to enable the effective transmission of information and data from the storage device **354** to the controller **360** and/or the remote device **320**. Alternatively, the information and data that is stored within the storage device **354** can be transmitted to the controller **360** without the need for a separate transmitter **356**. For example, the data can be transmitted via a removable cord to a computer or other processor.

(118) The receiver **358** can be utilized to receive any information and data that may be transmitted by the remote device **320**, such as the height and weight of the user. The receiver **358** can have any suitable design to enable the effective reception of information and data from the remote device **320**, which can subsequently be transmitted to and stored within the storage device **354**.

(119) The controller **360** is electrically coupled to the one or more performance sensors **316P** and/or the one or more locational sensors **316L**, such as via the storage device **354** and/or the transmitter **356**. The performance characteristics that are measured and/or sensed by the one or more performance sensors **316P** and/or the one or more locational sensors **316L** can be subsequently transmitted to and received by the controller **360**, and transmitted to and received by the remote device **320**, for conversion into usable statistical data, such as into one or more usable statistical data points. In one embodiment, one or more wires (not illustrated) can be utilized for transmitting the performance characteristics from the one or more performance sensors **316P** and/or the one or more locational sensors **316L** to the controller **360** and/or the remote device **320**, such as via the storage device **354** and/or the transmitter **356**. Alternatively, in another embodiment, the one or more performance sensors **316P** and/or the one or more locational sensors **316L** can be wirelessly coupled to the controller **360** and/or the remote device **320** for transmission of such performance characteristics.

(120) As noted, the controller **360** can be utilized to process and/or convert the performance characteristics as measured and/or sensed by the sensors **316P**, **316L** into usable statistical data for the athlete. Such statistical data can further incorporate the data input by the athlete via the remote device **320** (or otherwise), and/or such statistical data can be provided independent of the data input by the athlete. The controller **360** can include one or more circuits and/or processors. In many embodiments, the controller **360** can include one or more program algorithms that can be effectively utilized to convert the information from the sensors **316P**, **316L** into the desired usable statistical data. In many embodiments, the remote device **320** can include a lace adjuster application **380** that can include one or more program algorithms that can be effectively utilized to convert the information from the sensors **316P**, **316L** into the desired usable statistical data. The program algorithms can be varied depending on the particular statistical data that is desired.

(121) As noted above, the sensor assembly **316** can be utilized to generate various types of usable statistical data to gauge the performance of the athlete. For example, the sensor assembly **316** can be utilized to generate statistical data relating to substantially horizontal movements of the athlete, such as number of steps taken, total distance traveled, distance traveled per step (or stride length),

stride duration, ground contact duration and pattern, foot acceleration and foot angle during strides (gait tracking), speed of travel, and/or horizontal burst (such as sudden acceleration from an average rate of speed). Stride length can obviously vary depending on the nature of the specific activity. For example, when you are tired or running uphill you have shorter strides, and when you are fresh and/or running downhill you have longer strides. By averaging such information, and comparing that to the nature of the course to be run, the user can use this information to estimate how long it will take to finish the run. With the addition of the real-time clock **352**, this data can be further analyzed to generate statistical data for the horizontal speed of travel.

(122) In certain embodiments, statistical data with regard to horizontal burst can be generated by comparing the performance characteristics that have been measured and/or sensed by two-axis accelerometers (measuring acceleration along the X axis or the Y axis, as well as the Z axis) to the performance characteristics as measured and/or sensed by the three-axis accelerometer (measuring acceleration along each of the X axis, the Y axis and the Z axis). By subtracting the two-axis data from the three-axis data, the acceleration data for the off-axis can be determined. By so isolating the acceleration data along the X axis and along the Y axis, the horizontal burst can be effectively determined. As noted above, horizontal burst can be defined as sudden acceleration from an average rate of speed (whether the athlete is already moving or is at a dead stop). Such burst can further be defined from any directional vector, north, south, east, west, and anywhere in between. Burst algorithms need to average the force expended or acceleration rate, over time.

(123) In a substantially similar manner, the performance characteristics from the one or more sensors **316P**, **316L** can be utilized to generate statistical data regarding substantially vertical movements of the athlete, such as a number of jumps (once characteristics of what constitutes a jump are effectively established), height of jumps, jump duration and/or vertical burst (such as take-off velocity or acceleration for a jump). For example, in order to effectively determine what may constitute a jump and the height of the jump, information from a two-axis accelerometer (such as along the X axis and the Y axis) would be compared to the three-axis accelerometer, so that off axis movement (or non-true movement of the foot, when calculating height) can be removed from the analysis.

(124) The statistical data for the substantially horizontal movements of the athlete and for the substantially vertical movements of the athlete can be combined to generate additional desired statistical data, such as an overall number of accelerations (horizontal and vertical). The number of accelerations can be defined from zero momentum, to different monitoring of constant speed or a constant g range. Subsequently, a sudden increase in speed in any direction can be effectively quantified. Such information can be more valuable in certain sports that rely more on constant accelerations, such as ice hockey, or basketball.

(125) The three-dimensional gyrometer or other rate sensor can be utilized to analyze angular, twisting, or rotational movements of the athlete. In such analysis, it may be necessary to quantify how many degrees of angular movement or rotation from true will quantify as a twist and or rotation.

(126) The performance characteristics that are measured and/or sensed by the one or more sensors **316P**, **316L** can be further utilized to generate statistical data in relation to energy expended during athletic performance (such as in kcal), and/or force expended during athletic performance (such as in psi, kpi, or other force measurements). It should be appreciated that any statistical data related to energy expended and/or force expended can require information such as the weight of the athlete in order for such statistical data to be accurately generated.

(127) The power source **362** can provide the necessary power to the one or more performance sensors **316P**, the one or more locational sensors **316L**, the real-time clock **352**, the storage device **354**, the transmitter **356**, the receiver **358** and/or the controller **360** to enable all of these components to perform their desired functions. In one embodiment, the power source **362** can include one or more batteries (not shown), such as rechargeable batteries and/or single-use

batteries, which can be used to provide such necessary power. Alternatively, the power source **362** can have another suitable design.

(128) The remote device **320** can have any suitable design for interacting with and sending data and information to and receiving data and information from the sensor assembly **316** that is coupled to the lace adjuster **14**. For example, the remote device **320** can be a smartphone, a smart watch, a tablet, a computer, and/or any other suitable computing device. As illustrated in FIG. **3**, the remote device **320** can include one or more of a device body **364**, a connector port **366** that is formed into the device body **364**, an input mechanism **368**, a transmitter **370** (illustrated as a box in phantom), a receiver **372** (illustrated as a box in phantom), a storage device **374** (illustrated as a box in phantom), a display screen **376**, a controller **378** (illustrated as a box in phantom), a lace adjuster application **380** (illustrated as a box in phantom), and a power source **382**. Alternatively, the remote device **320** can have another suitable design, which can comprise more or fewer components than those specifically illustrated in FIG. **3**.

(129) As shown, in one embodiment, the device body **364** can provide a housing for the input mechanism **368**, the transmitter **370**, the receiver **372**, the storage device **374**, the display screen **376**, the controller **378**, the lace adjuster application **380**, and the power source **382**. The design of the device body **364** can be varied and/or the device body **364** can have any suitable shape.

(130) In certain embodiments, the device body **364** can include the connector port **366**, such as a USB port or other suitable connection, that enables the user to simply and directly connect the lace adjuster **14** to the remote device **320** to quickly and easily download any and all data generated through use of the sensor assembly **316**. With such design, the user is able to view any and all such data at a later time of convenience to the user.

(131) The input mechanism **368** provides a means by which the user can input any desired information into the feedback assembly **15** (illustrated in FIG. **1**) and/or the sensor assembly **216** to aid in deriving the desired statistical data. For example, in certain implementations, the input mechanism **368** can be utilized by the user to input information such as the height and weight of the user, which can subsequently be used for purposes of deriving statistical data related to energy expended and force expended during an athletic performance. Additionally, or in the alternative, the input mechanism **368** can be utilized by the user to input additional information.

(132) The transmitter **370** can be utilized to transmit any desired information and data from the remote device **320** to the receiver **358** of the sensor assembly **316**. For example, the transmitter **370** can be utilized to transmit any derived statistical data, as well as any data and information that was input through the input mechanism **368** from the remote device **320** to the sensor assembly **316**. The transmitter **370** can have any suitable design to enable the effective transmission of information and data from the remote device **320** to the sensor assembly **316**.

(133) The receiver **372** is configured to receive any data and information, such as any performance characteristics that have been sensed by the one or more performance sensors **316P** and/or the one or more locational sensors **316L**, that is transmitted from the sensor assembly **316** to the remote device **320**. The receiver **372** can have any suitable design for purposes of receiving the data and information from the sensor assembly **316**.

(134) The storage device **374** is utilized to store any data and information that is derived and/or utilized within the sensor assembly **316**. For example, the storage device **374** can be utilized to store any data and information that has been transmitted from the sensor assembly **316** to the remote device **320**. In some embodiments, the storage device **374** can further be utilized to store any statistical data that has been derived from the data and information that has been transmitted from the sensor assembly **316** to the remote device **320**. In certain embodiments, the storage device **374** can also be used to store any data and information that has been input by the user via the input mechanism **368**.

(135) The display screen **376** can be a video screen, of any suitable size and shape, which is utilized to display any and all data and information that is sensed, input and/or generated within the

sensor assembly **316**. More specifically, the display screen **376** can be utilized to display any performance characteristics that are measured and/or sensed by the one or more performance sensors **316P** and/or the one or more locational sensors **316L**, and data or information that is input by the athlete via the input mechanism **368** (or otherwise), and any statistical data points that may be generated from the sensed and input data by the controller **378**.

(136) The controller **378** is electrically coupled to the one or more performance sensors **316P** and/or the one or more locational sensors **316L**, such as via the storage device **374** and/or the receiver **372** incorporated into the remote device **320**, as well as the storage device **354** and/or the transmitter **356** incorporated within the assembly body **350** of the sensor assembly **316**. In certain embodiments, the performance characteristics that are measured and/or sensed by the one or more performance sensors **316P** and/or the one or more locational sensors **316L** are subsequently transmitted to and received by the controller **378** for conversion into usable statistical data, such as into one or more usable statistical data points. The controller **378** can include one or more processors or circuits for providing such functionality.

(137) As shown in FIG. 3, in one embodiment, the lace adjuster application **380** can be incorporated into the controller **378** for enabling the user to easily and consistently use the feedback assembly **15** (and/or the feedback system when the user has separate lace adjuster assemblies **13** coupled to each shoe **10A**, **10B**) that is built into the lace adjuster assembly **13**. The controller **378** and/or the lace adjuster application **380** can include an algorithm that is specifically configured to enable the derivation of any and all desired statistical data based on the data and information that is received within the controller **378** and which can be accessed by the lace adjuster application **380**.

(138) In some embodiments, the lace adjuster application **380** can be usable so that any desired data and information can be uploaded to a website for analysis, comparison, storage, or other suitable purposes.

(139) In certain embodiments, the remote device **320** will have Bluetooth capabilities, and have a social media aspect where customers can communicate and compare their statistical data to one or more professional athletes. It should be appreciated that this comparison of statistical data can embody many different sports.

(140) The power source **382** can provide the necessary power to the remote device **320** for enabling the desired functionality. More specifically, the power source **382** can provide the necessary power to each of the input mechanism **368**, the transmitter **370**, the receiver **372**, the storage device **374**, the display screen **376**, the controller **378** and the lace adjuster application **380** for purposes of enabling the desired functionality. In one embodiment, the power source **382** can include one or more batteries, such as rechargeable batteries and/or single-use batteries, which can be used to provide such necessary power. Alternatively, the power source **382** can have another suitable design.

(141) FIG. 4 is a simplified schematic illustration of an embodiment of the image assembly **417**. FIG. 4 further includes a simplified schematic illustration of an embodiment of the remote device **420** that can be utilized in conjunction with and/or as part of the image assembly **417**. As shown in FIG. 4, the remote device **420** can be substantially identical to, or actually the same remote device **320** that is utilized in conjunction with and/or as part of the sensor assembly **316** illustrated in FIG. 3. This image assembly **417** can be used in each of the lace adjuster assemblies **13** illustrated in FIG. 1.

(142) The design of the image assembly **417** can be varied. For example, as illustrated in FIG. 4, the image assembly **417** can be a digital camera that includes an assembly body **484**, an optical assembly **486**, a capturing system **488** (illustrated in phantom), a storage device **490** (illustrated as a box in phantom), a transmitter **492** (illustrated as a box in phantom), a controller **494** (illustrated as a box in phantom), and a power source **496** (illustrated as a box in phantom). The design of these components can be varied to suit the design requirements and type of image assembly **417**.

Alternatively, the image assembly **417** can be designed without one or more of these components. (143) In certain alternative embodiments, the image assembly **417** can be designed to capture still images of the athlete during an athletic performance, and/or the image assembly **417** can be designed to capture video image sequences of the athlete during an athletic performance. In some embodiments, the image assembly **417** can be activated manually by the athlete or other user of the image assembly **417**, and/or the image assembly **417** can be designed to be automatically activated based on the occurrence of certain movements or events.

(144) As shown in this embodiment, each of the optical assembly **486**, the capturing system **488**, the storage device **490**, the transmitter **492**, the controller **494** and the power source **496** can be coupled to and/or positioned substantially within the assembly body **484**. Alternatively, one or more of the components can be provided remotely from the assembly body **484**.

(145) The assembly body **484** can be rigid and support and/or provide a housing for at least some of the other components of the image assembly **417**, such as the optical assembly **486**, the capturing system **488**, the storage device **490**, the transmitter **492**, the controller **494** and the power source **496**. In one embodiment, the assembly body **484** includes a generally rectangular-shaped hollow body that forms a cavity that receives and retains such components of the image assembly **417**. Alternatively, the assembly body **484** can have another suitable shape.

(146) The optical assembly **486** can include a single lens or a combination of lenses that work in conjunction with each other to focus light onto the capturing system **488**. As the image assembly **417** is coupled to the lace adjuster **14** (illustrated in FIG. **1**), the optical assembly **486** can be positioned and oriented such that the lenses focus light onto the capturing system **488** from any desired direction. For example, in one embodiment, the optical assembly **486** can be positioned and oriented such that the lenses focus light onto the capturing system **488** from a generally vertical direction, such as when the optical assembly **486** is directed in a generally upward direction from the lace adjuster **14**. Additionally, and/or alternatively, the optical assembly **486** can be positioned and oriented such that the lenses focus light onto the capturing system **488** from a generally horizontal direction and/or at any desired angle between the vertical and horizontal directions.

(147) In one embodiment, the image assembly **417** includes an autofocus assembly (not shown) including one or more lens movers that move one or more lenses of the optical assembly **486** in or out until the sharpest possible image of a main subject, such as the athlete, is received by the capturing system **488**.

(148) The capturing system **488** captures information for the still images and/or the video sequences of the athlete during their athletic performance. The design of the capturing system **488** can vary according to the type of image assembly **417**. For a digital-type camera, the capturing system **488** can include an image sensor (not shown) and a filter assembly (not shown).

(149) The still images and/or video sequences that are captured by the capturing system **488** can be stored and/or maintained within the storage device **490** of the image assembly **417**. The storage device **490** can have any suitable design that enables the storing of such still images and/or video sequences.

(150) The transmitter **492** can be utilized to transmit the still images and/or video sequences that are stored within the storage device **490** to the controller **494** and/or to the remote device **420**, such as a television, a smart phone, a computer, etc. The transmitter **492** can have any suitable design to enable the effective transmission of the still images and/or video sequences from the storage device **490** to the controller **494** and/or to the remote device **420**. Alternatively, the still images and/or video sequences that are stored within the storage device **490** can be transmitted to the controller **494** without the need for a separate transmitter **492**.

(151) The controller **494** is electrically connected to and controls the operation of the electrical components of the image assembly **417**. The controller **494** can include one or more processors and circuits, and the controller **494** can be programmed to perform one or more of the functions described herein. For example, the controller **494** can be utilized to perform various processing

steps on the still images and/or video sequences of the athlete that have been captured by the capturing system **488**.

(152) As shown, the controller **494** can be positioned within the assembly body **484**. In some embodiments, the controller **494** and/or a separate, second controller **478** can be positioned remotely from the image assembly **417**, such as within the remote device **420**.

(153) The power source **496** can provide the necessary power to the optical assembly **486**, the capturing system **488**, the storage device **490**, the transmitter **492** and/or the controller **494** to enable all of these components to perform their desired functions. In one embodiment, the power source **496** can include one or more batteries, such as rechargeable batteries and/or single-use batteries, which can be used to provide such necessary power. Alternatively, the power source **496** can have another suitable design.

(154) It should be appreciated that in embodiments of the lace adjuster assembly **13** (illustrated in FIG. **1**) that include both the image assembly **417** and the sensor assembly **316**, the transmitter **492**, the controller **494** and/or the power source **496** can be used in common for each of the image assembly **417** and the sensor assembly **316**. Alternatively, in such embodiments, the image assembly **417** and the sensor assembly **316** can include and utilize separate transmitters, controllers and/or power sources.

(155) In one embodiment of the lace adjuster assembly **13** that includes both the image assembly **417** and the sensor assembly **316**, the various components of the image assembly **417** and the sensor assembly **316** can be coupled to and/or positioned substantially within a common assembly body.

(156) In some embodiments, as shown in FIG. **4** and as noted above, the image assembly **417** can be wirelessly coupled to the remote device **420**. For example, in certain embodiments, the transmitter **492** of the image assembly **417** can be designed to wirelessly transmit the still images and video sequences of the athlete to the remote device **420** via Wi-Fi, Bluetooth, or other suitable wireless technique.

(157) The design of the remote device **420** can be varied. As noted above, the remote device **420** can be substantially identical to, or actually the same remote device **320** that is utilized in conjunction with and/or as part of the sensor assembly **316** illustrated in FIG. **3**. In particular, as illustrated, the remote device **420** can again include one or more of a device body **464**, a connector port **466** that is formed into the device body **464**, an input mechanism **468**, a transmitter **470** (illustrated as a box in phantom), a receiver **472** (illustrated as a box in phantom), a storage device **474** (illustrated as a box in phantom), a display screen **476**, a controller **478** (illustrated as a box in phantom), a lace adjuster application **480** (illustrated as a box in phantom), and a power source **482**. Each of the noted components have essentially the same design and functionality as described in detail herein above in relation to FIG. **3**.

(158) However, the use and application of the various components can be modified slightly for purposes of interacting as desired with the image assembly **417**. For example, the data and information being transmitted from the image assembly **417** to the remote device **420** is somewhat different than the data and information being transmitted between the sensor assembly **316** and the remote device **320**. More particularly, the image assembly **417** is configured to capture any desired images, such as still images and/or video images, that can be subsequently transmitted to the remote device **420**. Such images can then be stored in the storage device **474** and/or displayed on the display screen **476** as desired.

(159) FIG. **5** is a simplified top view of an area **598** that is usable by a user of the lace adjuster assembly **13** (illustrated in FIG. **1**). As noted above, in some embodiments, the lace adjuster assembly **13** can include a feedback assembly **15** (illustrated in FIG. **1**) that can include a locational sensor **216L** (illustrated in FIG. **2F**) such as a GPS sensor within the sensor assembly **216** (illustrated in FIG. **2F**) in order to provide locational and/or tracking information for the user. It is appreciated that in order to obtain the most precise and accurate locational and tracking

information, it can be desired to include more than one or more additional sensors **599**, such as two, three, or four sensors, that are spaced apart from one another and positioned near the area **598**. In such arrangement, the overall system is better able to determine and track the actual precise location of the lace adjuster assembly **13**. It is further appreciated that if the user wears a separate lace adjuster assembly **13** on each shoe **10A**, **10B** (illustrated in FIG. **1**), then the locational and tracking information provided by the feedback system can be even more precise and accurate.

(160) The type of area **598** can vary. For example, the area **598** can be an athletic field such as a football or soccer field; a court such as a tennis or basketball court; or another type of area **598**.

(161) In the present design, the user of the lace adjuster assembly **13** can be participating in an event in the area **598**, and the one or more additional sensors **599** can be used to improve the locational information of the lace adjuster assembly **13**.

(162) In the embodiment shown in FIG. **5**, the area **598** can include two additional sensors **599**. For example, as provided above, the lace adjuster assembly **13** and/or the locational sensors **216L** of the sensor assembly **216** can include a GPS sensor. Each additional sensor **599** can also include a GPS sensor (or GPS beacon) to determine their precise location relative to the area **598** and relative to the locational sensors **216L** of the lace adjuster assembly **13**. The GPS information from these additional sensor(s) **599** can be relayed to the lace adjuster assembly **13** and/or to the remote device **220** (illustrated in FIG. **2F**) to improve the measurement information of the lace adjuster assembly **13**. As non-exclusive examples, the lace adjuster assembly **13** can be electrically connected via WI-FI or Bluetooth to the additional sensors **599**.

(163) In some embodiments, the additional sensors **599** can be used to monitor the relative position of the lace adjuster assembly **13** over time. For example, the additional sensors **599** can include one or more systems that monitor the relative position of the lace adjuster assembly **13** over time, or generate signals that can be used by the lace adjuster assembly **13** to monitor position.

(164) In certain embodiments, the additional sensors **599** can generate GPS signals which can be utilized by the lace adjuster assembly **13** to provide more accurate and precise locational and tracking information for the user of the lace adjuster assembly **13**. It is appreciated that the additional sensors **599** can be positioned in any suitable manner relative to the area **598**, such as on and/or near the area **598**, in order to provide such information for the user. As shown, the additional sensors **599** will typically be provided in fixed positions relative to the area **598**. Thus, during use, each of the additional sensors **599** can provide precise, locational and/or tracking information. The additional sensors **599** can also be electronically linked to one another and/or can communicate with one another, such as wirelessly or with a wired connection.

(165) With this design, each of the additional sensor(s) **599** can communicate, such as wirelessly, in any suitable manner, with the locational sensor **216L**, such as the GPS sensor, within the sensor assembly **216** and/or the feedback assembly **15** as the user moves on or about the area **598**. Based on the communications among the additional sensor(s) **599** and the locational sensor **216L** within the sensor assembly **216** and/or the feedback assembly **15** of the lace adjuster assembly **13** on the shoelace **12** (illustrated in FIG. **1**) of the shoes **10A**, **10B** of the user, precise locational and/or tracking information of the user can be known at all times when the user is using the area **598**. In such manner, the user can obtain desired information regarding statistical data, such as during an athletic performance, in order to effectively gauge various aspects of their athletic performance.

(166) It is understood that although a number of different embodiments of the lace adjuster assembly **13**, the lace adjuster **14** and the feedback assembly **15** have been illustrated and described herein, one or more features of any one embodiment can be combined with one or more features of one or more of the other embodiments, provided that such combination satisfies the intent of the present invention.

(167) While a number of exemplary aspects and embodiments of the lace adjuster assembly **12**, the lace adjuster **14** and the feedback assembly **15** have been shown and disclosed herein above, those of skill in the art will recognize certain modifications, permutations, additions, and sub-

combinations thereof. It is therefore intended that the lace adjuster assembly **12**, the lace adjuster **14** and the feedback assembly **15** shall be interpreted to include all such modifications, permutations, additions, and sub-combinations as are within their true spirit and scope, and no limitations are intended to the details of construction or design herein shown.

Claims

1. A lace adjuster system that is adapted to selectively adjust a first shoelace of a first shoe of a user and to selectively adjust a second shoelace of a second shoe of the user, the lace adjuster system comprising: a first lace adjuster assembly including (i) a first lace adjuster that is adapted to selectively adjust the first shoelace of the first shoe of the user; and (ii) a first feedback assembly that is mechanically coupled to the first lace adjuster, the first feedback assembly measuring statistical data of the user during an athletic performance, the first feedback assembly including a first sensor assembly including a first performance sensor that is mechanically coupled to the first lace adjuster and that senses a first performance characteristic of the user during the athletic performance; and a first controller that is electrically coupled to the first performance sensor, the first controller including a first processor, the first controller receiving the first performance characteristic from the first performance sensor and generating a first statistical data point that is based at least in part on the first performance characteristic; and a second lace adjuster assembly including (i) a second lace adjuster that is adapted to selectively adjust the second shoelace of the second shoe of the user; and (ii) a second feedback assembly that is mechanically coupled to the second lace adjuster, the second feedback assembly measuring statistical data of the user during the athletic performance, the second feedback assembly including a second sensor assembly including a second performance sensor that is mechanically coupled to the second lace adjuster and that senses a second performance characteristic of the user during the athletic performance; and a second controller that is electrically coupled to the second performance sensor, the second controller including a second processor, the second controller receiving the second performance characteristic from the second performance sensor and generating a second statistical data point that is based at least in part on the second performance characteristic.
2. The lace adjuster system of claim 1 wherein the first controller is the same as the second controller.
3. The lace adjuster of claim 1 wherein the first controller and the second controller are each within a remote device.
4. The lace adjuster system of claim 1 wherein the first statistical data point and the second statistical data point are combined by one of the first controller and the second controller to generate a combined statistical data point having enhanced accuracy relative to the first statistical data point and the second statistical data point.
5. The lace adjuster system of claim 4 wherein the first feedback assembly further includes a first storage device for storing the combined statistical data point.
6. The lace adjuster system of claim 5 wherein the first storage device is mechanically coupled to the first lace adjuster; and wherein the first sensor assembly further includes a first transmitter for transmitting the combined statistical data point from the first storage device to a remote device.
7. The lace adjuster system of claim 6 wherein the first transmitter transmits the combined statistical data point from the first storage device to the remote device via a wireless connection.
8. The lace adjuster system of claim 6 wherein the first transmitter transmits the combined statistical data point from the first storage device to the remote device via a wired connection.
9. The lace adjuster system of claim 1 wherein the first performance sensor senses one or more of a horizontal movement, a vertical movement and an angular movement of the user during the athletic performance.
10. The lace adjuster system of claim 9 wherein the first performance sensor is one of a two-axis

accelerometer, a three-axis accelerometer, and a three-axis gyrometer.

11. The lace adjuster system of claim 9 wherein the first performance sensor includes a first magnetometer that measures a magnitude and direction of magnetic fields at a point in space in relation to a position of the user during the athletic performance.

12. The lace adjuster system of claim 9 wherein the second performance sensor senses the one or more of a horizontal movement, a vertical movement and an angular movement of the user during the athletic performance.

13. The lace adjuster system of claim 12 wherein the second performance sensor is the one of a two-axis accelerometer, a three-axis accelerometer, and a three-axis gyrometer.

14. The lace adjuster system of claim 12 wherein the second performance sensor includes a second magnetometer that measures a magnitude and direction of magnetic fields at a point in space in relation to a position of the user during the athletic performance.

15. The lace adjuster system of claim 1 wherein the first feedback assembly further includes a first image capturing assembly that captures a first image of the user during the athletic performance.

16. The lace adjuster system of claim 15 wherein the second feedback assembly further includes a second image capturing assembly that captures a second image of the user during the athletic performance.

17. The lace adjuster system of claim 1 wherein the first sensor assembly further includes a first locational sensor for providing precise locational information of the user; and wherein the locational information from the first locational sensor is wirelessly transmitted to a remote device.

18. The lace adjuster system of claim 17 wherein the second sensor assembly further includes a second locational sensor for providing precise locational information of the user; and wherein the locational information from the second locational sensor is wirelessly transmitted to the remote device.

19. The lace adjuster system of claim 1 wherein the first lace adjuster includes (i) a body assembly having a first body member and a second body member that is coupled to the first body member, the body assembly defining a cavity, and (ii) a lace end retainer that is connected to the body assembly, the lace end retainer being configured to selectively retain at least a portion of the shoelace, the lace adjuster being selectively movable between an unlocked configuration and a locked configuration, wherein the shoelace is adjustable relative to the lace adjuster when the lace adjuster is in the unlocked configuration, and wherein the shoelace is resiliently retained by the lace adjuster and is inhibited from being adjusted relative to the lace adjuster when the lace adjuster is in the locked configuration; and wherein the first performance sensor is positioned within the cavity.

20. A lace adjuster system that is adapted to selectively adjust a first shoelace of a first shoe of a user and to selectively adjust a second shoelace of a second shoe of the user, the lace adjuster system comprising: a first lace adjuster assembly including (i) a first lace adjuster that is adapted to selectively adjust the first shoelace of the first shoe of the user; and (ii) a first feedback assembly that is mechanically coupled to the first lace adjuster, the first feedback assembly measuring statistical data of the user during an athletic performance, the first feedback assembly including a first sensor assembly including a first performance sensor that is mechanically coupled to the first lace adjuster and that senses a first performance characteristic of the user during the athletic performance; and a first controller that is electrically coupled to the first performance sensor, the first controller including a first processor, the first controller receiving the first performance characteristic from the first performance sensor and generating a first statistical data point that is based at least in part on the first performance characteristic; and a second lace adjuster assembly including (i) a second lace adjuster that is adapted to selectively adjust the second shoelace of the second shoe of the user; and (ii) a second feedback assembly that is mechanically coupled to the second lace adjuster, the second feedback assembly measuring statistical data of the user during the athletic performance, the second feedback assembly including a second sensor assembly including

a second performance sensor that is mechanically coupled to the second lace adjuster and that senses a second performance characteristic of the user during the athletic performance; and a second controller that is electrically coupled to the second performance sensor, the second controller including a second processor, the second controller receiving the second performance characteristic from the second performance sensor and generating a second statistical data point that is based at least in part on the second performance characteristic; wherein the first statistical data point and the second statistical data point are combined by one of the first controller and the second controller to generate a combined statistical data point having enhanced accuracy relative to the first statistical data point and the second statistical data point; wherein the first feedback assembly further includes a first storage device for storing the combined statistical data point; wherein the first storage device is mechanically coupled to the first lace adjuster, the first sensor assembly further including a first transmitter for transmitting the combined statistical data point from the first storage device to a remote device; wherein the first performance sensor senses one or more of a horizontal movement, a vertical movement and an angular movement of the user during the athletic performance; wherein the second performance sensor senses the one or more of a horizontal movement, a vertical movement and an angular movement of the user during the athletic performance; wherein the first feedback assembly includes a first image capturing assembly that captures a first image of the user during the athletic performance; wherein the second feedback assembly includes a second image capturing assembly that captures a second image of the user during the athletic performance; wherein the first sensor assembly further includes a first locational sensor for providing precise locational information of the user, the locational information from the first locational sensor being wirelessly transmitted to a remote device; and wherein the second sensor assembly further includes a second locational sensor for providing precise locational information of the user, the locational information from the second locational sensor being wirelessly transmitted to the remote device.
